

Road and Off road Vehicles Report

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1 Ride Comfort Laboratory

1.1 Introduction

The evolution of vehicle dynamics has been driven by the continuous search of an optimal balance between conflicting performance indices: ride comfort, handling, and safety. This report focuses on the analysis of vertical dynamics to understand how vehicles have evolved from the past to the present days in terms of ride comfort and which are the most influent design parameters when assessing the vertical dynamics performances of a vehicle.

1.2 Case of Study:

Three representative vehicles from the same market segment (Alfa Romeo) which belongs to different automotive eras have been selected to conduct the study:

- **Alfa Romeo Giulietta (1960)**
- **Alfa Romeo Alfetta (1984)**
- **Alfa Romeo 159 (2007)**

The vehicle vertical dynamics is analyzed using a Quarter Car Model, which represents a simplified 2 DOF lumped-parameter system corresponding to a single corner of the vehicle.

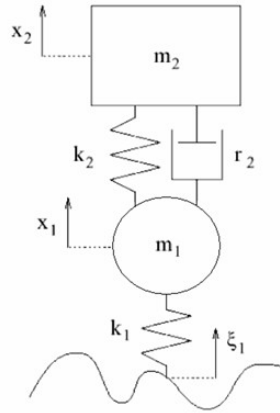


Figure 1: Quarter Car Model Scheme

The parameters are:

- m_1 : unsprung mass [kg];
- m_2 : sprung mass [kg];
- k_1 : tire radial stiffness [N/m];
- k_2 : suspension equivalent stiffness [N/m];
- r_2 : suspension equivalent damping [Ns/m];
- ξ_1 : road irregularity [m];

	Giulietta	Alfetta	Alfa 159
m1	25 kg	30 kg	33 kg
m2	200 kg	270 kg	350 kg
k1	100000 N/m	150000 N/m	270000 N/m
k2	10000 N/m	20000 N/m	28000 N/m
r2	1500 Ns/m	2500 Ns/m	5200 Ns/m

1.3 Frequency Response Analysis

The dynamic behavior of the three vehicles is first analyzed in the frequency domain.

1.3.1 Natural Frequencies Identification:

The Quarter Car Model is a 2 DOF system, characterized by two distinct natural frequencies; these can be analytically approximated as follows:

- **Body Bounce Frequency (f_{n1}):** It is the low-frequency mode where the sprung mass (m_2) oscillates the most:

$$f_{n1} \approx \frac{1}{2\pi} \sqrt{\frac{k_1 k_2}{m_2(k_1 + k_2)}}$$

- **Wheel Hop Frequency (f_{n2}):** It is the high-frequency mode where the unsprung mass (m_1) oscillates the most:

$$f_{n2} \approx \frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m_1}}$$

	Giulietta	Alfetta	Alfa 159
f_{n1}	1.07 Hz	1.28 Hz	1.35 Hz
f_{n2}	10.55 Hz	11.98 Hz	15.12 Hz

It can be noticed that for all the vehicles, the suspension parameters tuning is always done to prevent the natural frequencies to fall inside the 4 - 8 Hz range which is considered as dangerous for humans as it coincides with some body resonant frequencies.

1.3.2 Frequency Response Functions (FRF)

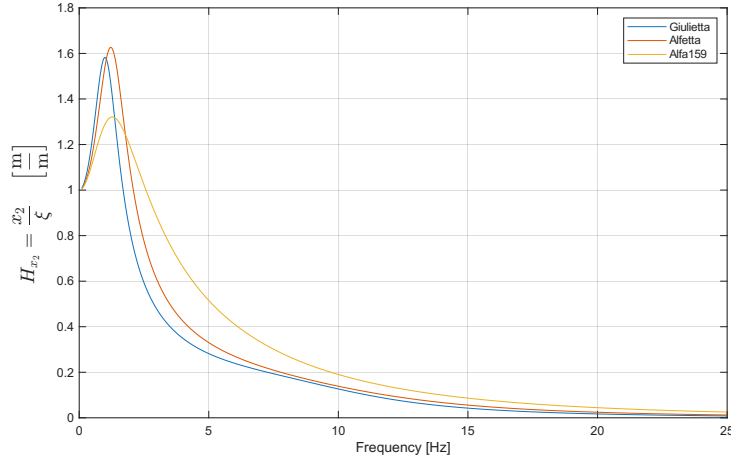


Figure 2: $|H_{x_2}| = \frac{x_2}{\xi}$

The Frequency Response Function (FRF) magnitude between ξ and x_2 shows how much the first resonance mode is felt by the sprung mass, while the second peak is much more attenuated if not even imperceptible, due to the damping effect of the suspension.

The **Giulietta** shows a more pronounced peak at the lower frequency compared to the 159. This indicates a "softer" setup. On the contrary, the **Alfa 159** shows a highly damped response; the high value of r_2 (5200 Ns/m) significantly reduces the magnitude of the resonant peak, improving body control but implying a stiffer connection to the road.

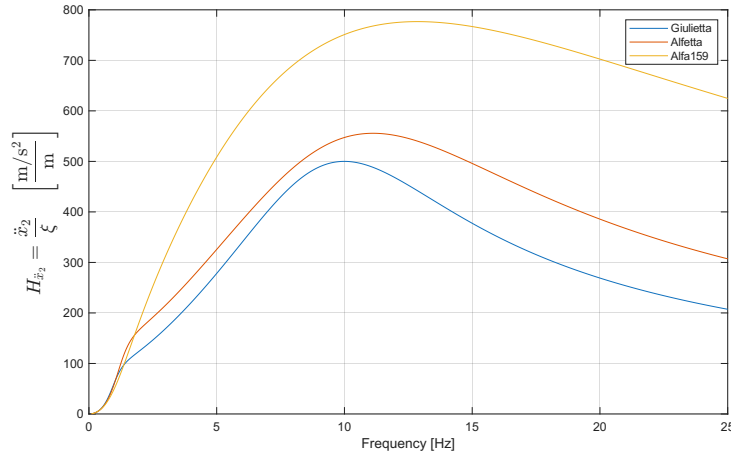


Figure 3: $|H_{\ddot{x}_2}| = \frac{\ddot{x}_2}{\xi}$

The Frequency Response Function (FRF) magnitude between ξ and the sprung mass vertical acceleration $\ddot{x}_2$ is computed as:

$$H_{\ddot{x}_2}(\omega) = -\omega^2 \cdot H_{x_2}(\omega)$$

Physically, $|H_{\ddot{x}_2}(\omega)|$ represents the magnification factor of the vertical acceleration experienced by the passenger relative to a road amplitude $\xi = 1$; it describes the intensity of the inertial forces acting on the human body. Its increasing trend with frequency reflects the physical reality that rapid variations in road profile generate significant inertial forces.

1.3.3 Power Spectral Densities (PSD)

The Power Spectral Density of $\ddot{x}_2$ can be computed as:

$$S_{\ddot{x}_2}(\omega) = |H_{\ddot{x}_2}(\omega)|^2 \cdot S_{\xi}(\omega)$$

where $S_{\xi} = \frac{AbV}{\omega^2}$ is the Road Power Spectral Density. For this study there were two possible road configurations:

	Normal Road	Rough Road
Ab	1.4e-5 m	1.0e-4 m

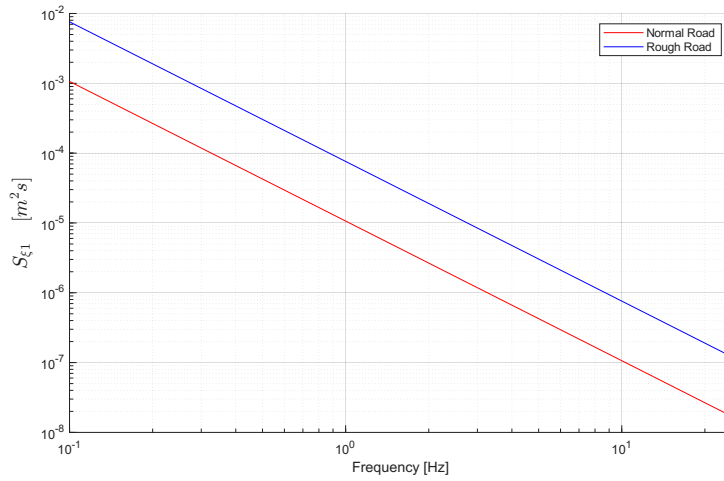


Figure 4: S_{ξ}

The Road brings lots of energy at low frequencies (i.e long wavelengths, associated with large up and downs) while for higher frequencies the energy is drastically reduced (i.e short wavelengths, associated with small defects and rugosity of the asphalt).

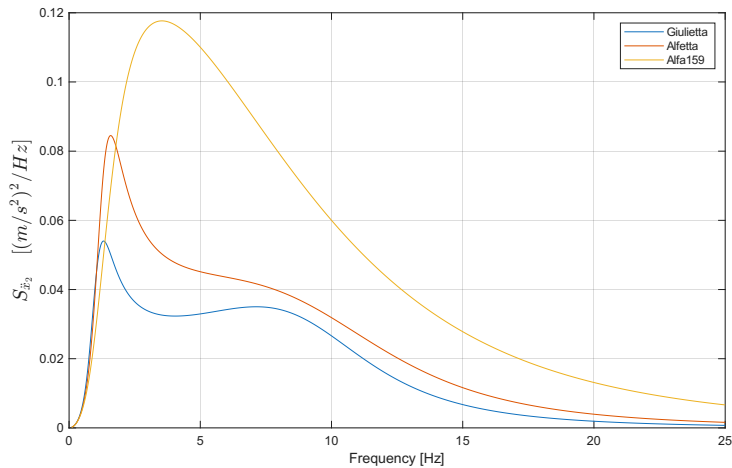


Figure 5: $S_{\ddot{x}_2}$

Since the road excitation spectrum decays with frequency ($S_{\xi} \propto \omega^{-2}$), the input energy at high frequencies is low. This decay counteracts the ω^2 amplification of the acceleration transfer function. Consequently, in the final PSD $S_{\ddot{x}_2}$, the low-frequency Body

Bounce peak regains its importance, indicating that despite the vehicle's high sensitivity to high-frequency acceleration, the main energy content of the vibration experienced by the passenger comes from road irregularities in the lower frequency range.

The area under these curves represents the total energy of the vibration; visually, the Giulietta appears to have less energy content compared to the stiffer vehicles

1.4 Discomfort Index Evaluation

To quantify the ride quality, the Discomfort Index ($\sigma_{\ddot{x}_2}$) is calculated. The definition of the Discomfort Index is related to the $\ddot{x}_2$ RMS evaluation:

$$\sigma_{\ddot{x}_2} = \sqrt{\frac{1}{2\pi} \int_{-\infty}^{+\infty} |H_{\ddot{x}_2}|^2 \cdot S_{\xi} d\omega}$$

The calculation is performed using the analytical solution for a Quarter Car model:

$$\sigma_{\ddot{x}_2} = \sqrt{\frac{A_b V}{2} \left[\frac{(m_1 + m_2) k_2^2}{m_2^2 r_2} + \frac{k_1 r_2}{m_2^2} \right]}$$

The indices for the three vehicles are evaluated at a reference speed of $V = 30$ m/s (108 km/h) on the "Normal Road" profile ($A_b = 1.4e - 5$ m):

	Giulietta	Alfetta	Alfa 159
$\sigma_{\ddot{x}_2}$	0.93 m/s^2	1.10 m/s^2	1.58 m/s^2

1.4.1 Sensitivity Analysis

To identify which design parameters have the most significant impact on comfort, a sensitivity analysis is conducted on the Quarter Car model.

The analysis follows the **One-At-a-Time (OAT)** approach: starting from a nominal configuration (Baseline), individual parameters were varied within a defined range while keeping the others constant. The following result shows the Discomfort Index ($\sigma_{\ddot{x}_2}$) trend due to the influence of each single parameter:

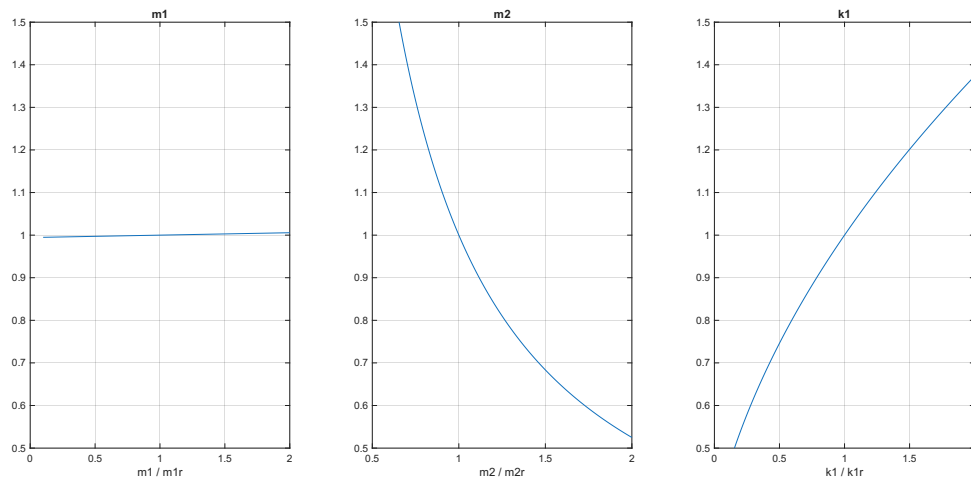


Figure 6

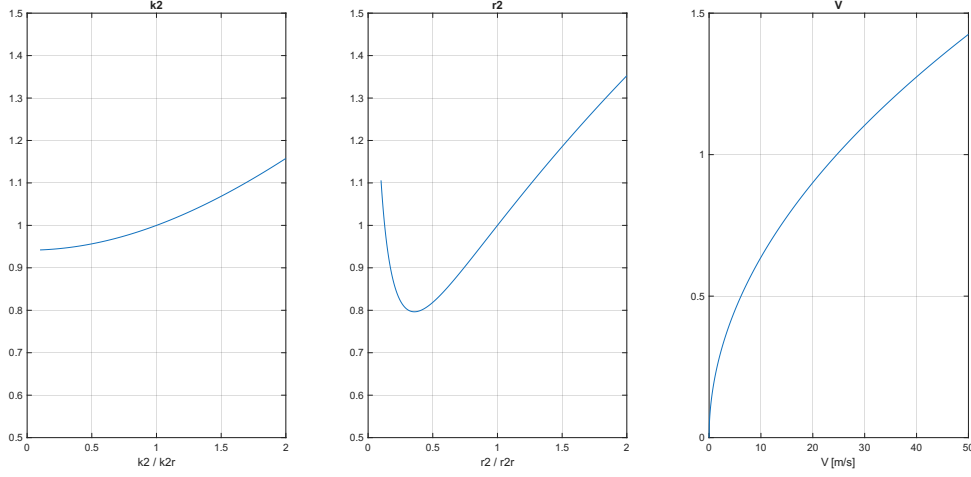


Figure 7

- m_1 : the effect of the unsprung mass is secondary compared to other parameters, but a positive trend is present. This is due to the inertial effect of the wheel at high frequencies, which transfers more forces to the chassis if m_1 is high.
- m_2 : a strong inverse sensibility is observed. A high inertial mass acts as a natural low-pass filter, effectively opposing the sudden motion variations imposed by the forces transmitted by the suspension. This explains why heavier vehicles are intrinsically more stable and comfortable.
- k_1 : a strong positive trend is observed; the tire acts as a "primary spring" and controls the Wheel Resonance. A high k_1 value stiffens the connection between the road profile and the unsprung mass, facilitating the transmission of high-frequency irregularities.
- k_2 : a positive trend is observed; a stiffer spring transmits larger forces to the vehicles body and makes it more "nervous" to drive.
- r_2 : the sensitivity to suspension damping shows a unique behavior. If r_2 is high, the suspension becomes more rigid and the decoupling between the sprung and unsprung masses becomes weaker resulting in a more "harsh" suspension which is not able to filter the motion of the wheel. On the contrary, a too low value of r_2 makes the suspension too soft which makes the resonant peaks not damped enough. We can see from the sensitivity plot that exists a trade-off value for r_2 which is able to optimize the Discomfort.
- V : A strictly monotonic positive trend is observed. Increasing the speed has two effects: it increases the amplitude of the road excitation input power (more energy enters the system per unit time) and it shifts the road input spectrum towards higher frequencies.

To minimize completely the Discomfort, we would need theoretically to:

- k_2 , k_1 and $r_2 \rightarrow 0$: without any springing and damping effect, there would not be possible to transmit vertical forces to the chassis, the decoupling effect between the chassis and the tires would be maximized.

- $m_2 \rightarrow \infty$: if the sprung mass (chassis) tends to infinite, it would be impossible to move it vertically, hence its vertical acceleration $\ddot{x}_2$ would be null.

Of course, this would be practically impossible to obtain and would lead to a vehicle with poor handling (Lateral Dynamics) capability.

Moreover, we may also consider the presence of other indexes which define the vertical performance of a suspension system, like the Working Space ($\sigma_{x_2-x_1}$) and the Road Holding (σ_{F_z}): due to their conflicting nature, a trade-off design choice must be done.

1.4.2 Vehicles Comparison:

The **Giulietta** achieves the lowest Discomfort value, confirming its status as the most comfortable vehicle in terms of pure vertical isolation. This is primarily due to the low suspension stiffness (k_2) and the relatively low damping (r_2), which minimize the transmission of forces.

- **Giulietta vs Alfetta:** Theoretically, the Alfetta ($m_2 = 270$ kg) starts with an advantage over the Giulietta ($m_2 = 200$ kg) due to its higher inertia. However, the Alfetta features a suspension stiffness ($k_2 = 20000$ N/m) that is double of the Giulietta ($k_2 = 10000$ N/m). The negative impact of the stiffer suspension on the Discomfort Index overpowers the positive benefit provided by the extra mass.
- **Giulietta vs Alfa 159:** The Alfa 159 is significantly heavier ($m_2 = 350$ kg), which would naturally lead to better Comfort. However, to manage this mass and ensure safety/handling, the suspension is stiffened drastically ($k_2 = 28000$ N/m) and the damping is increased more than threefold ($r_2 = 5200$ Ns/m)
- **Old vs Modern Suspension Design Choices:** the analysis leads to a counter-intuitive conclusion which suggests that older vehicles are more comfortable than modern ones. The main reason is related to how new vehicles are designed and tuned according to modern standards, in order to get a good compromise between all the performance indexes, able to ensure drivability but also safety.
 - **Handling Capability:** modern vehicles are required to ensure lateral stability also at high speeds; a stiffer setup of the suspensions limits the body roll and guarantees stability and good steering response during emergency maneuvers. This improves the overall safety.
 - **Mass of the vehicle:** nowadays vehicles are way more heavier due to safety equipments and comfort features, To support this mass (m_2) without hitting the suspension bump stops, stiffer springs (k_2) are required. High damping (r_2) is then necessary to dissipate the energy of the heavy oscillating mass, preventing the vehicle from bouncing uncontrollably after a bump.
 - **Mass of the tires:** nowadays tires are larger and heavier, their inertia must be taken into account whenever we want them to maintain contact with the road: a stiffer suspension setup is needed to keep the tire attached to the ground
 - **First frequency mode position:** frequencies below 1 Hz are the primary cause of the so called "motion sickness" felt by the driver. By shifting the natural frequency slightly higher, modern vehicles are able to prevent the "seasick" effect, trading a bit of harshness for better physiological well-being.

1.5 Simulation on Asphalt Road with Cleats

In order to better visualize the different behavior of the three vehicles, a riding test equal to the one showed in the lectures is performed via Simulink using a Quarter Car model. The test involves the vehicle traveling at a speed of $V = 20\text{km/h}$ on a road with some cleats spaced equally apart:

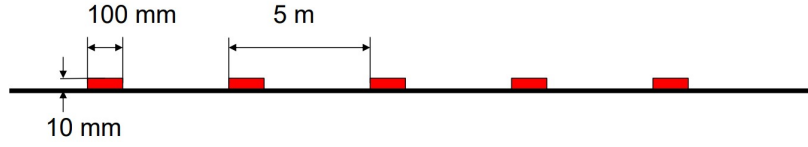


Figure 8: Cleats Sequence

1.5.1 Road Modeling

The time-domain simulations were performed using a hybrid road profile model. The total vertical road displacement excitation, $\xi(t)$, is modeled as the superposition of two distinct contributions:

$$\xi_{tot}(t) = \xi_{roughness}(t) + \xi_{cleats}(t)$$

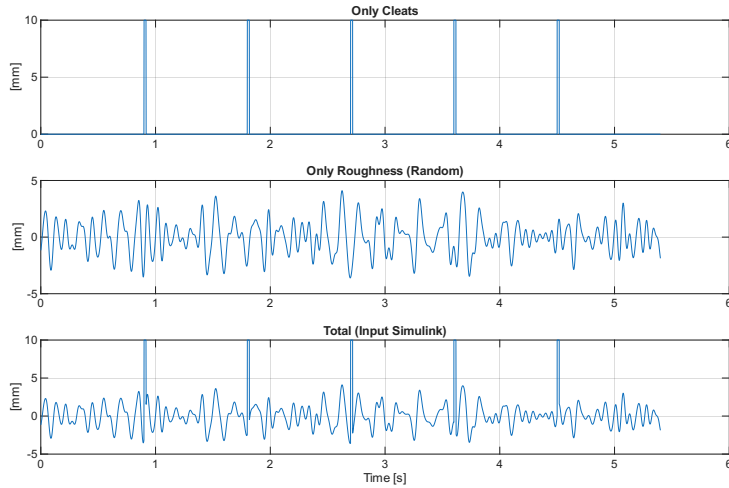


Figure 9: Road Profile

- $\xi_{roughness}(t)$: the road irregularity is modeled as a stationary Ergodic Gaussian Stochastic process: the transition from the spectral domain (PSD) to the time domain was achieved using the **Harmonic Superposition Method**. This approach models the road profile as a finite sum of N discrete sinusoidal waves, each characterized by a specific frequency, amplitude, and random phase.

$$\xi_{roughness}(t) = \sum_{k=1}^N A_k \cdot \sin(\omega_k t + \phi_k)$$

For the k -th sinusoidal amplitude A_k evaluation, we rely on the principle of energy conservation: the energy generated by the signal within a frequency band $\Delta\omega$ must be equal to the area under the target PSD curve $S_\xi(\omega)$ for that same band.

For a single sinusoid with amplitude A_k , the power is given by $A_k^2/2$. By equating this to the spectral power:

$$\frac{A_k^2}{2} = S_\xi(\omega_k) \cdot \Delta\omega \quad \Rightarrow \quad A_k = \sqrt{2 \cdot S_\xi(\omega_k) \cdot \Delta\omega}$$

- $\xi_{cleats}(t)$: Superimposed on the random roughness is a deterministic sequence of discrete obstacles. These are modeled as a series of rectangular pulses intended to simulate harsh impacts.

1.5.2 Results

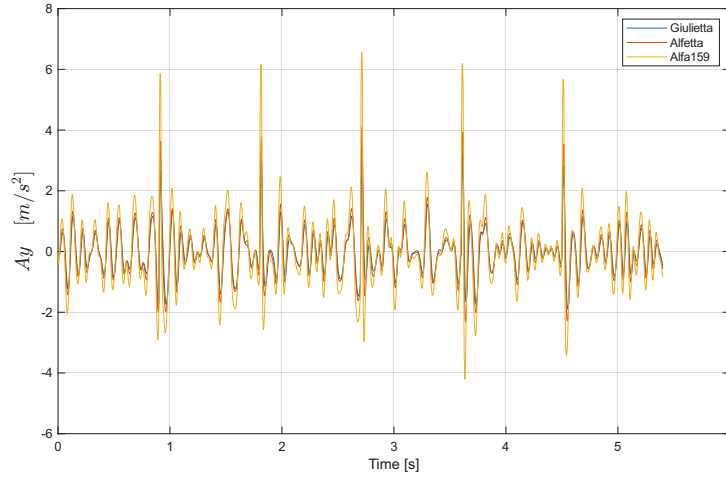


Figure 10: A_y Vertical Body Acceleration

The graph confirms that the **Giulietta** seems to be the best vehicle in terms of vertical comfort since it's able to filter out the road irregularities and the acceleration spikes due to the impact with the cleats more than the other models.

2 Vehicle Performance Laboratory

2.1 Introduction

The primary objective of this laboratory is to evaluate the longitudinal dynamic performance of a passenger vehicle, with a specific focus on determining its maximum speed and acceleration capabilities. A critical prerequisite for assessing performance is the accurate construction of the engine torque and power curves at the wheels. This requires a detailed design of the transmission system, specifically the selection of the appropriate gear ratios.

2.2 Case of Study

The vehicle selected for this analysis is the **Aston Martin DB5**, a classic British Grand Tourer.



Figure 11: Aston Martin DB5

Technical specifications used for the simulation have been derived from available literature and specialized automotive databases.

Geometry	
Front Area	$1.680 \cdot 1.340 \text{ m}^2$
Length	4.570 m
WheelBase	2.490 m
Track (Front)	1.372 m
Track (Rear)	1.359 m
Tires	
R_0 (Radius)	0.3465 m
k_{tire}	200000 N/m
$\Delta R = \frac{mg}{4k_{tire}}$	0.0196 m
$R = R_0 - \Delta R$	0.3269 m

Mass and Inertial Terms	
$m_{vehicle}$	1465 kg
m_{driver}	75 kg
Tank Capacity	86 l
J_m (Motor)	$0.28 \text{ kg} \cdot \text{m}^2$
J_w (Wheel)	$0.55 \text{ kg} \cdot \text{m}^2$
J_t (Transm.)	$0.07 \text{ kg} \cdot \text{m}^2$

Since specific experimental data for the rotational inertia were not available, parameters like J_m , J_w and J_t were estimated based on literature data provided for vehicle classes.

The values were selected by identifying the vehicle category that best approximates the characteristics of the chosen Grand Tourer.

The vehicle is powered by a 4.0-liter Inline-6 naturally aspirated engine. The powertrain includes a ZF 5-speed manual gearbox.

Gear ratios $\tau_{ci} [\omega_{out}/\omega_{in}]$	
I	0.364
II	0.568
III	0.811
IV	1
V	1.199
Final Drive Ratio $[\omega_{out}/\omega_{in}]$	
τ_p	0.279

Table 1: Transmission ratios from manufacturer

The figure below shows the Engine Power and Torque Maps provided by the manufacturer:

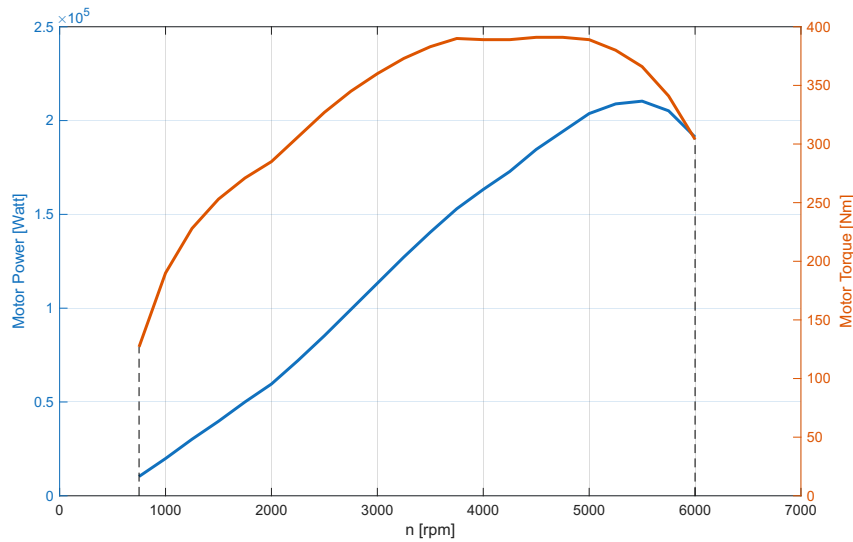


Figure 12: Engine Power and Torque Maps

Top Speed	234km/h at 5700rpm
Max Torque	391Nm at 4500rpm
Max Power	286cv at 5500rpm

2.3 Power and Resistant Curves

Although the vehicle is a complex multi-body system, for the purpose of this longitudinal dynamics analysis, it is modeled as a point-mass on which all the forces are applied.

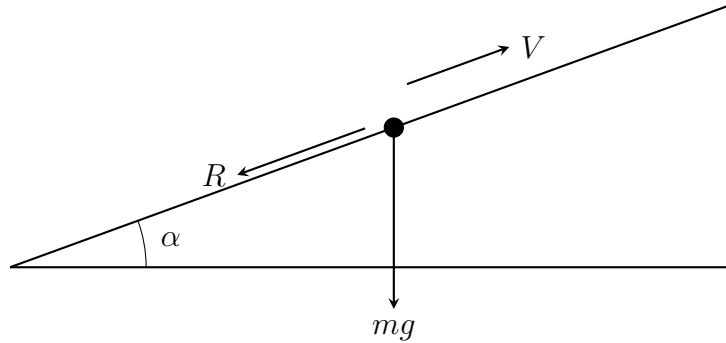


Figure 13: Point-mass vehicle model

The lumped mass of the vehicle (m) is moving at constant speed (V) on a straight path with a ground inclination (α). For the design phase, the total mass of the vehicle considers also the presence of the driver and a 90% full fuel tank.

The resistant forces are defined as:

$$R = R_{slope} + R_{roll} + R_{Aero}$$

- **Resistance due to road slope:** $R_{slope} = mg \cdot \sin(\alpha)$
- **Rolling resistance:** $R_{roll} = mg \cdot \cos(\alpha) \cdot (f_0 + f_2 V^2)$

f_0	0.01 (Smooth asphalt)
f_2	$0.5 \cdot 3.6^2 \cdot 10^{-6}$

- **Aerodynamic resistance:** $R_{Aero} = \frac{1}{2} \rho_{air} C_x S V^2$

ρ_{air}	1.225 kg/m^3
C_x	0.45
$S = \text{Front Area} \cdot \text{Fill Factor}$	$1.680 \cdot 1.340 \cdot 0.9 \text{ m}^2$

The Engine and Resistance power curves as function of the vehicle speed are reported below: to be comparable, we need to consider the available power at the wheels provided by the engine. To do so, the following conversion is done considering possible power losses due to the components' efficiency:

$$W_{mot,wheels} = \eta_C \cdot \eta_D \cdot \eta_P \cdot W_{mot,engine}$$

- $\eta_C = 0.97$: Gear box efficiency of the i -th gear.
This assumption is justified by the mechanical characteristics of the vehicle's manual transmission, which employs helical gears. According to the provided technical reference, helical gears operate within an efficiency range of $[0.94, 0.98]$, making 0.97 a representative value for a high-performance gearbox.

- $\eta_D = 1$: Transaxle efficiency. Since the Aston Martin DB5 is a Rear Wheel Driven (RWD) vehicle, the drive axle is only one, thus the transaxle efficiency is unitary.
- $\eta_P = 0.96$: Final drive efficiency (Differential efficiency).
The differential efficiency is obtained considering the case when the vehicle is reaching its top speed; this represents a well detailed situation since the top speed and the corresponding rpm are provided by the manufacturer.

$$\begin{cases} W_{mot,wheels} = W_{res}(V_{max}, \alpha = 0^\circ) \\ W_{mot,wheels} = \eta_C \cdot \eta_D \cdot \eta_P \cdot W_{mot,engine}(rpm = 5700) \end{cases}$$

$$\Rightarrow \eta_P = \frac{W_{res}(V_{max}, \alpha = 0^\circ)}{\eta_C \cdot \eta_D \cdot W_{mot,engine}(rpm = 5700)}$$

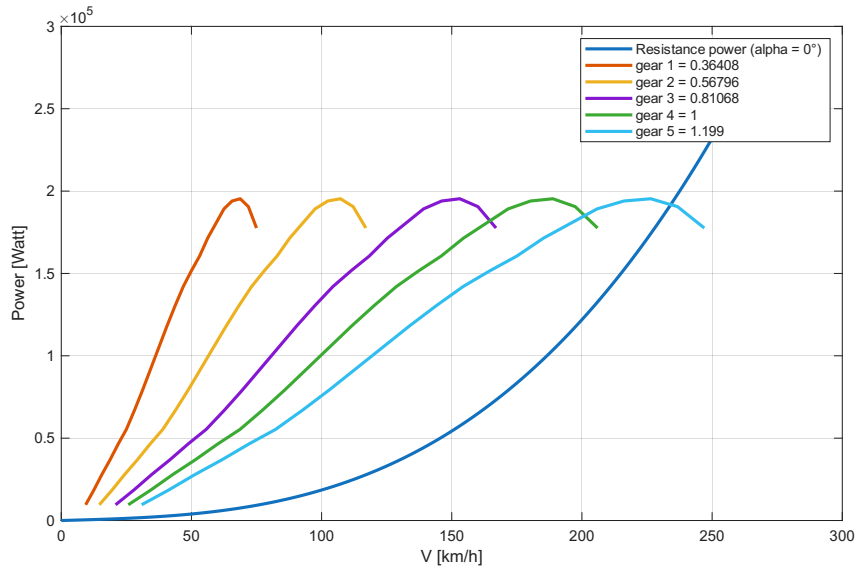


Figure 14: Resistant curve and Available Power at the wheels considering the gear set provided by the manufacturer

2.4 Gear Set Design

The gear ratios set are designed differently with respect to the ones provided by the manufacturer. The highest and lowest gear ratios are selected according to some directives, while the intermediate ones are obtained through a geometric progression formula.

2.4.1 Highest Gear Ratio

The highest gear ratio (τ_{max} , i.e the 5th gear) is determined with the objective of achieving the maximum vehicle speed on flat ground ($\alpha = 0^\circ$). This ratio is calculated by varying the transmission ratio value until the curve of the available power at the wheels ($W_{mot,wheels}$) intersects the curve of the resistant power (W_{res}) at the desired top speed.

The selection also depends on the desired balance between performance and efficiency. According to vehicle dynamics theory, there are three possible design strategies regarding the intersection point:

1. **Maximum Speed Design:** The ratio is chosen so that the intersection occurs exactly at the engine speed corresponding to maximum power. Ideally, this allows the vehicle to reach the highest possible velocity.
2. **Maximum Acceleration Design:** The ratio is "shorter", causing the intersection to occur at an engine speed higher than the peak power rpm. In this configuration, the engine operates at higher rpms, ensuring a larger reserve of power is available for acceleration, at the cost of a slightly lower top speed.
3. **Fuel Economy Design:** The ratio is "longer", causing the intersection to occur at an engine speed lower than the peak power RPM. This forces the engine to operate at lower angular speeds during highway cruising (where high speeds are reached), reducing friction losses and fuel consumption, although the vehicle will not be able to reach its theoretical top speed.

In our case, the maximum gear ratio is selected according to the "Maximum Speed Design" approach.

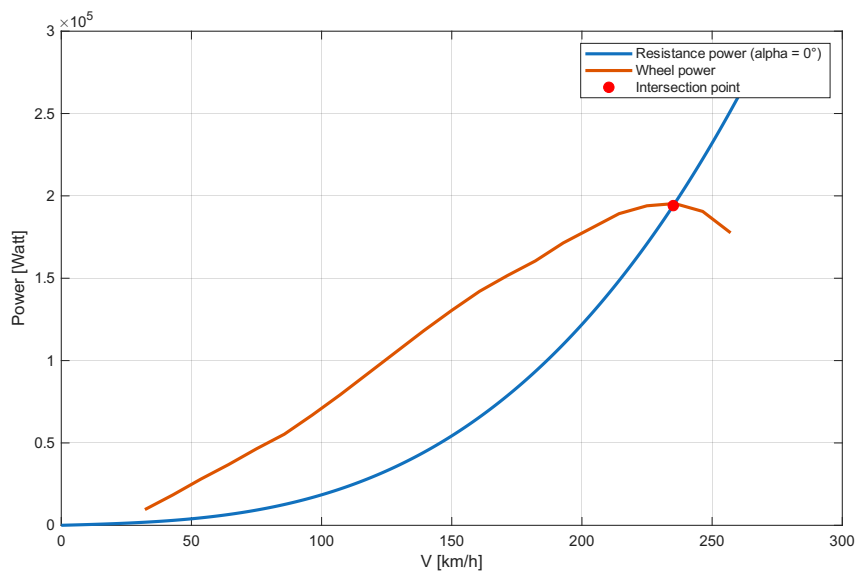


Figure 15: Resistant curve and Available Power at the wheels considering the designed gear set

2.4.2 Lowest Gear Ratio

The lowest gear ratio (τ_{min} , i.e the 1st gear) is selected to ensure the vehicle can overcome a maximum slope of 45%. To guarantee that the vehicle has sufficient residual power to accelerate, the ratio is determined by making the engine power curve tangent to the resistant power curve calculated at a gradient of 46%.

Furthermore, a kinematic check is performed to verify low-speed drivability: at the engine's minimum angular speed (approx. 800 – 1000 rpm), the vehicle speed must fall within the range of 7 – 12 km/h.

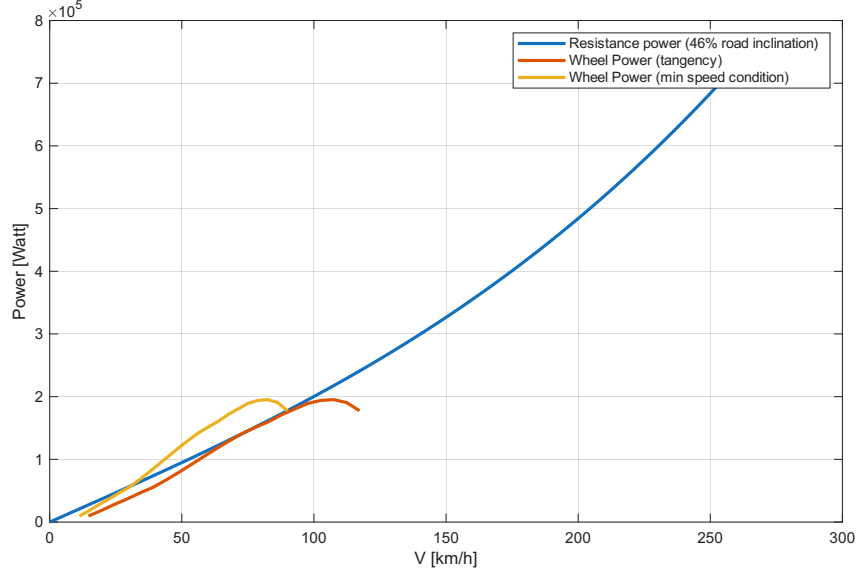


Figure 16: Lowest Gear Selection

The gear ratio that satisfy the tangency condition, fails the kinematic check since it is too high; the minimum gear ratio is selected in order to guarantee a speed of 12km/h at 800rpm.

2.4.3 Final Gear Set

The intermediate gear ratios are determined using a geometric progression: this approach ensures that the variation in engine angular speed remains constant when shifting from one ratio to the next, optimizing the engine's operating range.

$$\tau_t = \frac{\tau_{min}}{\tau_{max}}$$

The relationship between two subsequent gear ratios τ_{i-1} and τ_i is:

$$\frac{\tau_{i-1}}{\tau_i} = \tau_{tot}^{\frac{1}{n-1}}$$

where n represents the total number of gears.

The following gear set is provided:

Gear ratios $\tau_{ci} [\omega_{out}/\omega_{in}]$	
I	0.437
II	0.568
III	0.0.738
IV	0.960
V	1.248
Final Drive Ratio $[\omega_{out}/\omega_{in}]$	
τ_p	0.279

Table 2: Designed Transmission ratios

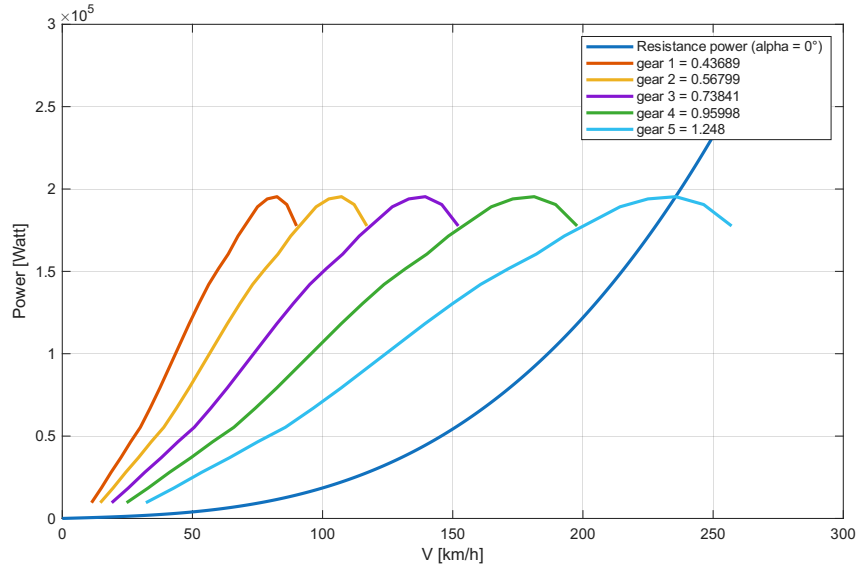


Figure 17: Resistant curve and Available Power at the wheels considering the designed gear set

2.5 Acceleration Maneuver

To test the final performance, an acceleration maneuver is performed in order to see how long does the vehicle take to go from 70 to 120km/h with the transmission ratio corresponding to the highest gear (τ_{max}). The acceleration time is calculated by discretizing the speed range: the time required to increase the velocity ΔV assuming constant acceleration is calculated as $\Delta t = \Delta V/a$. The useful power available for the vehicle to accelerate is given by the difference between the available engine power at the wheels and the resistant curve inside the speed range of interest:

$$W_{useful} = W_{mot,wheels} - W_{res}$$

The acceleration value is computed as:

$$a = \frac{W_{useful}}{V m_{eq}}$$

where m_{eq} is the Equivalent Mass, which takes into account also the rotational inertia terms from the wheels and the motor. m_{eq} can be derived from the energetic approach:

$$\begin{aligned} & \begin{cases} W_{useful} = \frac{dE_k}{dt} = m_{eq} a V \\ E_k = \frac{1}{2} m_{vehicle} V^2 + \frac{1}{2} (4 J_w \omega_w^2) + \frac{1}{2} J_m \omega_m^2 + \frac{1}{2} J_t \omega_t^2 \end{cases} \\ \Rightarrow \frac{dE_k}{dt} &= m_{vehicle} a V + 4 J_w \omega_w \dot{\omega}_w + J_m \omega_m \dot{\omega}_m + J_t \omega_t \dot{\omega}_t = m_{eq} a V \\ \Rightarrow m_{eq} &= m_{vehicle} + 4 \frac{J_w}{R^2} + \frac{J_m}{R^2 (\tau_{max} \tau_p)^2} + \frac{J_t}{R^2 \tau_p^2} \end{aligned}$$

The simulation are performed on two different loading situations:

- **Running load**

- $m_{driver} = 70$ kg
- $m_{luggage} = 5$ kg
- Half tank of fuel

- **Full load**

- $m_{driver} = 70$ kg
- $m_{passenges} = 3 \cdot 70$ kg (4 total seats)
- $m_{luggage} = 4 \cdot 15$ kg (15kg of luggage per person)
- Full tank of fuel

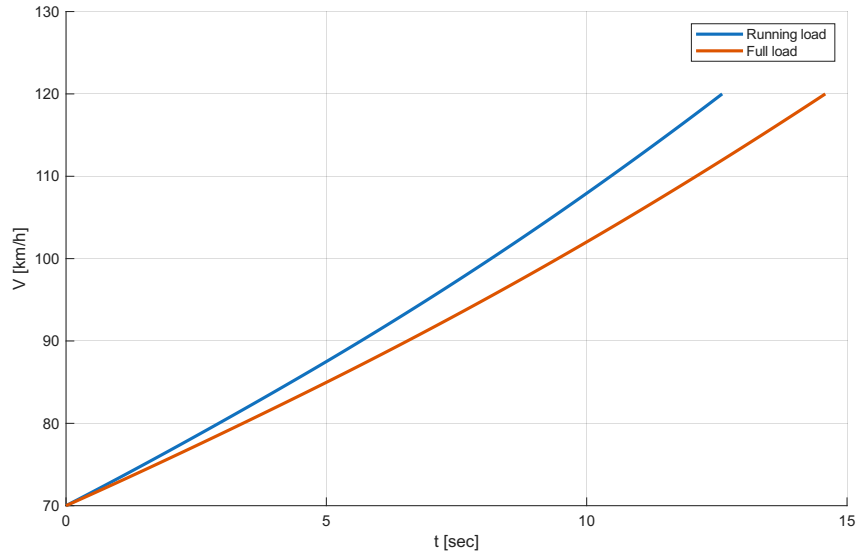


Figure 18: Vehicle Speed

The total acceleration time is then computed as:

$$\Delta t_{tot} = \sum_i \frac{\Delta V_i}{a}$$

Mode	Total Time
Running Load	12.61 sec
Full Load	14.59 sec

3 Vehicle Consumption Laboratory

3.1 Introduction

In the modern automotive landscape, the analysis of energy consumption and pollutant emissions plays an important role in vehicle development. With increasingly stringent environmental regulations and a growing market demand for efficiency, it is essential to accurately evaluate how a vehicle performs under realistic operating conditions.

To ensure consistency, vehicles must be tested according to standardized procedures regulated by international norms. These tests are designed to verify consumption across various driving scenarios, ranging from congested urban traffic to high-speed highway cruising.

This laboratory will specifically examine the application of these standards to a Plug-in Hybrid Electric Vehicle (PHEV), analyzing its behavior and energy management strategies during a standard driving cycle.

3.2 Case of study

3.2.1 Vehicle

The vehicle under analysis is a Plug-in Hybrid Electric Vehicle (PHEV) characterized by a "Double Drive" architecture. In this specific configuration, there is no link between the front and rear axles. Instead:

* Front Axle is powered by an internal combustion engine (ICE). The torque is transferred to the front wheels through a transmission and a front final drive. * Rear Axle is powered by an electric motor (EM). The EM draws energy from a battery pack via a power converter and drives the rear wheels through a dedicated rear final drive.

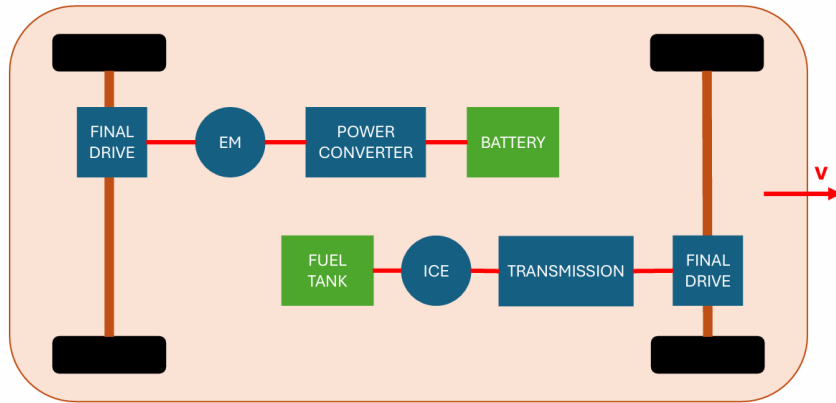


Figure 19: vehicle scheme

This layout allows the vehicle to operate with front-wheel drive (ICE only), rear-wheel drive (electric mode), or all-wheel drive when both ICE and EM are active simultaneously. The data of the vehicle are collected in the following table:

Table 3PHEV Technical Specifications.

Parameter	Value / Description
Weight (without driver)	1770 kg
Aerodynamic factor ($C_x \cdot S$)	0.913 m ²
Tyres	235/45 R19
Traction configuration	4 × 4 (FWD ICE, RWD Electric)
Internal Combustion Engine (ICE)	
Type	Spark-ignition (line 4, gasoline)
Displacement	1332 cm ³
Max Power	132 kW @ 5800 rpm
Max Torque	270 Nm @ 1000–6500 rpm
RPM range	1000–6500 rpm
Electric Motor (EM)	
Specs (Power/Torque)	45 kW / 250 Nm
Battery capacity	11.40 kWh
Regeneration efficiency	0.3
Transmission	
Number of gears	6
Gear ratios	4.459, 2.508, 1.555, 1.142, 0.851, 0.672
Final drive ratio (Front)	3.683
Overall drive ratio (Rear)	8

The initial approach for defining the ICE full-load performance involved the digital extraction of the upper boundary directly from the manufacturer's efficiency map. However, this method yielded discrepancies when compared to the official tabular data, specifically failing to accurately match the maximum rated power declared at 5800 RPM.

To resolve this inconsistency, an analytical approach was adopted. A fourth-order polynomial was utilized to model the engine's specific work (Brake Mean Effective Pressure, BMEP), which ensures a smooth and accurate reconstruction of the performance curves. To resolve this inconsistency, an analytical approach was adopted. A fourth-order polynomial ($y = an^4 + bn^3 + cn^2 + dn + e$) was utilized to model the engine's specific work (Brake Mean Effective Pressure, BMEP). The polynomial coefficients were calibrated by solving a linear system based on the following boundary conditions:

- **Starting point:** The initial specific work value at 1000 RPM was extracted directly from the torque and power curves provided in the assignment.
- **Maximum torque:** The peak specific work value was imposed at 2700 RPM.
- **Torque peak derivative:** The first derivative of the specific work curve was set to zero at 2700 RPM, establishing a local maximum.

- **Maximum power:** The specific work value at 5800 RPM was calculated to precisely match the declared maximum power output.
- **Power peak derivative:** Tangency to the constant power hyperbola was imposed at 5800 RPM. By setting the first derivative $y' = -y/n$, the power is mathematically constrained to reach its absolute maximum exactly at this engine speed.

The following MATLAB snippet illustrates the implementation of these conditions to construct the coefficient matrix:

```

1 % 1. DEFINITION OF KEY POINTS AND CONDITIONS
2 % Point 1: Start
3 n_start = 1000;
4 y_start = 1.621;      % Extracted from assignment curves
5
6 % Point 2: Max Torque
7 n_peak = 2700;
8 y_peak = 2.55;
9 dy_peak = 0;          % Local maximum
10
11 % Point 3: Max Power Limit
12 n_end = 5800;
13 y_end = 2.05;
14 dy_end = -y_end / n_end; % Tangency to constant power curve
15
16 % 2. LINEAR SYSTEM CONSTRUCTION (Ax = B)
17 % Polynomial: P(n) = a*n^4 + b*n^3 + c*n^2 + d*n + e
18 A = [
19     n_start^4,    n_start^3,    n_start^2,    n_start,    1;
20     n_peak^4,     n_peak^3,     n_peak^2,     n_peak,     1;
21     4*n_peak^3,   3*n_peak^2,   2*n_peak,     1,         0;
22     n_end^4,      n_end^3,      n_end^2,      n_end,      1;
23     4*n_end^3,    3*n_end^2,    2*n_end,      1,         0
24 ];

```

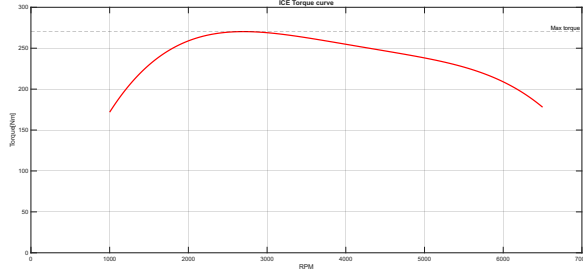
Once the polynomial for the specific work (L_{spec}) was defined, the full-load engine torque T_{ICE} was derived using the standard fundamental equation for four-stroke engines, which relates specific work to engine displacement (V_d):

$$T_{ICE} = \frac{L_{spec} \cdot V_d}{4\pi} \quad (1)$$

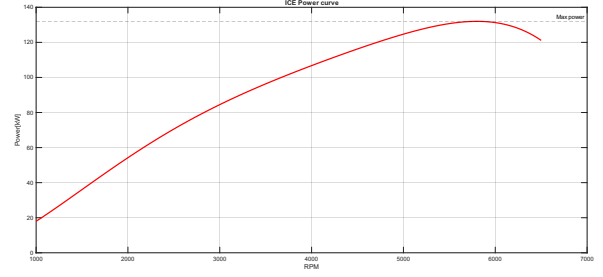
(Note: appropriate conversion factors are applied depending on the chosen units for pressure and volume). Subsequently, the mechanical power P_{ICE} was calculated using the standard rotational kinematic relationship:

$$P_{ICE} = T_{ICE} \cdot \omega \quad (2)$$

where ω represents the engine's angular velocity.



(a) ICE Torque Curve

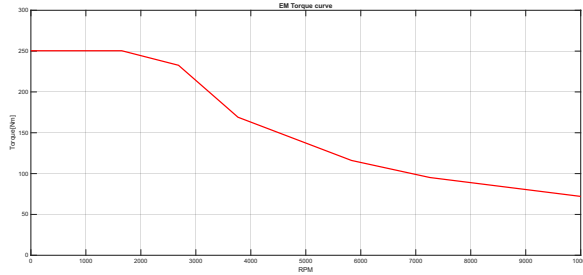


(b) ICE Power Curve

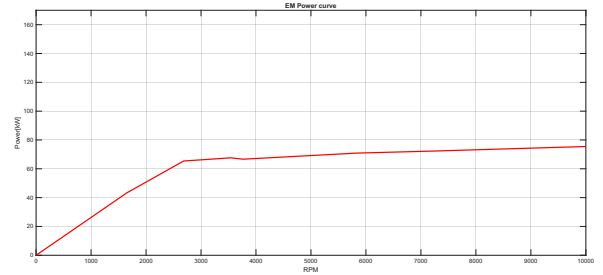
Figure 20 Internal Combustion Engine performance curves.

For the electric motor, the performance limits were consistent with the expected operational data. Therefore, the maximum torque curve was directly acquired via digital extraction from the upper boundary of the EM efficiency map. The mechanical power limit P_{EM} as a function of the motor speed was then derived through the direct algebraic conversion of the extracted torque T_{EM} :

$$P_{EM} = T_{EM} \cdot \left(\frac{\text{RPM} \cdot 2\pi}{60} \right) \quad (3)$$



(a) EM Torque Curve

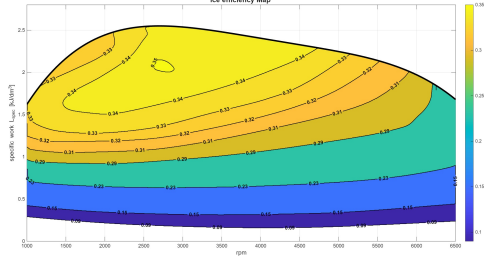


(b) EM Power Curve

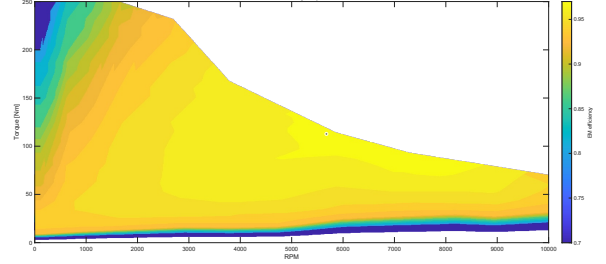
Figure 21 Electric motor performance curves.

To implement the efficiency models within the vehicle simulation, the raw contour lines from both the ICE and EM efficiency maps were digitally extracted. These discrete data points, representing constant efficiency boundaries (iso-efficiency curves), were processed to reconstruct a continuous operational domain.

A 2D spatial interpolation was performed over the entire engine speed and load range (torque or specific work). This process assigned a precise efficiency value to every possible operating point, resulting in high-resolution, color-coded matrices (lookup tables). This methodology provides robust, discrete zones that guarantee rapid and accurate efficiency evaluations during the WLTP cycle simulation.



(a) ICE Torque Curve



(b) ICE Power Curve

Figure 22 efficiency maps.

3.2.2 WLTP cycle

To perform this analysis, the reference standard used is the **Worldwide Harmonized Light Vehicles Test Procedure (WLTP)**.

The WLTP cycle is a global standard designed to determine the consumption and emissions of vehicles in a repeatable manner. The cycle covers a total duration of 1800 seconds (30 minutes).

Vehicles are classified according to their power to unladen mass ratio (PMR [$Watt/kg$]):

Class	PMR range [$Watt/kg$]
1	$PMR \leq 22$
2	$22 < PMR \leq 34$
3	$PMR > 34$

According to our vehicle's powertrain, a Class 3 WLTP cycle must be used assuming a top speed above 120 km/h.

The cycle is divided into four distinct phases that simulate a progressive increase in driving intensity, moving from city traffic to highway cruising:

- **Low Phase:** this section simulates urban driving. It is characterized by low average speeds and frequent stops, requiring constant acceleration and deceleration (simulating traffic lights or congestion).
- **Medium Phase:** represents suburban driving with higher speeds and fewer stops.
- **High Phase:** simulates rural or extra-urban driving scenarios.
- **Extra High Phase:** represents highway driving, where the vehicle reaches its highest speeds and maintains them for longer periods.

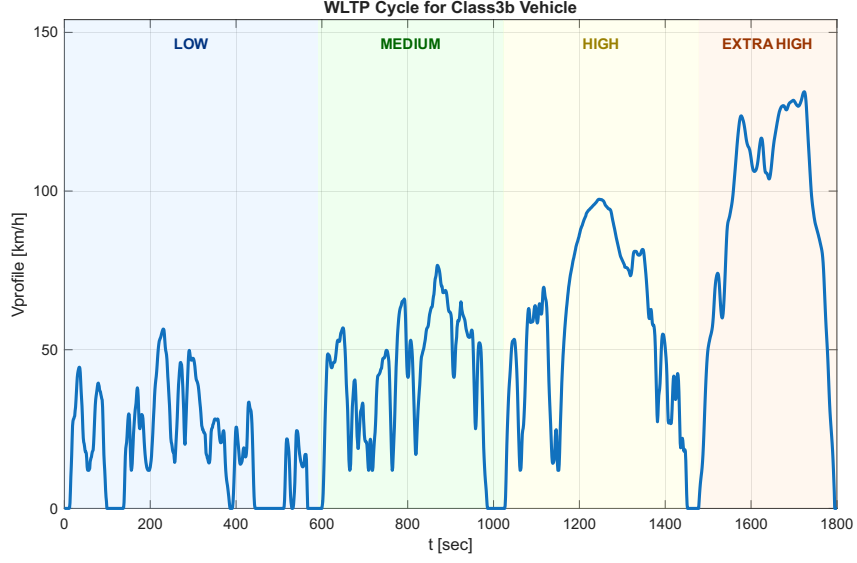


Figure 23: WLTP Cycle Speed Profile

3.2.3 Resistant Forces and Required Power

To evaluate the consumptions, the energy demand profile along time must be derived from the total resistance force (R_{tot}) opposing the vehicle's motion. For the simulation, an adopted total mass m includes both the curb mass of the vehicle and the presence of a driver of 75 kg:

$$R_{tot} = R_{Roll} + R_{Aero} + ma$$

- **Rolling resistance:** $R_{roll} = mg \cdot \cos(0^\circ) \cdot (f_0 + f_2 V^2)$

f_0	0.01 (Smooth asphalt)
f_2	$0.5 \cdot 3.6^2 \cdot 10^{-6}$

- **Aerodynamic resistance:** $R_{Aero} = \frac{1}{2} \rho_{air} C_x S V^2$

ρ_{air}	1.225 kg/m^3
$C_x \cdot S$	0.913 m^2

- **Inertial resistance:** ma , which must be taken into account due to the vehicle's continuous change of speed

The required power profile at the drive shaft P_{shaft} is computed as:

$$P_{shaft} = \frac{R_{tot} V}{\eta_t}$$

where $\eta_t = 0.95$ is the transmission efficiency.

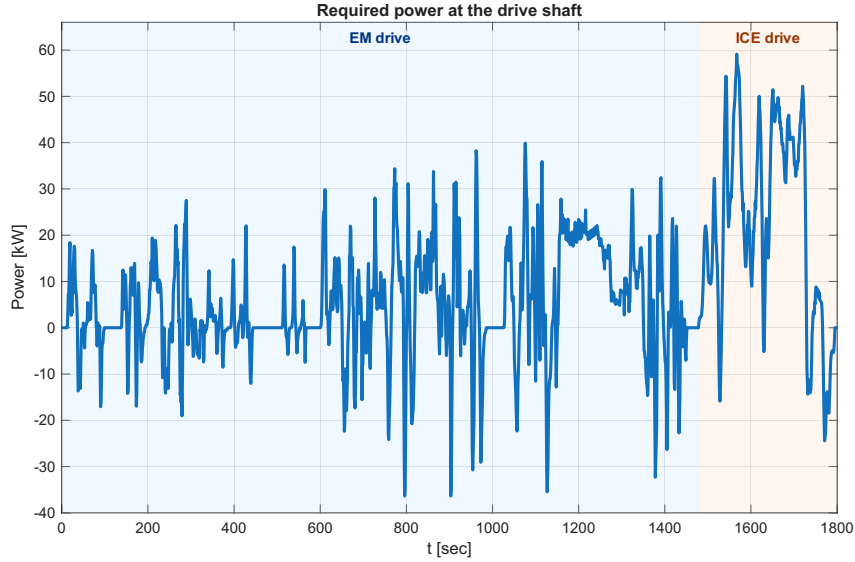


Figure 24: P_{shaft}

3.3 Simulation

The simulation divides the WLTP cycle into specific operating zones. This logic mimics the behavior of a sophisticated Energy Management System, which decides which power source to utilize based on vehicle speed and torque demand.

- **EM Drive:** this phase covers the Low, Medium and High phases of the cycle: the traction torque is delivered exclusively by the Electric Motor, while the Internal Combustion Engine (ICE) is switched off to maximize efficiency in stop-and-go traffic, eliminating fuel consumption and local emissions in the urban scenario.
- **ICE Drive:** this phase covers the Extra-High phase of the cycle: the traction is primarily provided by the ICE. This strategy leverages the engine's characteristics, as the ICE operates more efficiently at higher speeds compared to the electric motor.

3.3.1 EM Drive

The ICE is totally switched off, whenever the required power is positive, it is exclusively provided by the Electric Motor, on the other hand, during decelerations, the Electric Motor acts like a generator which is able to recharge the batteries.

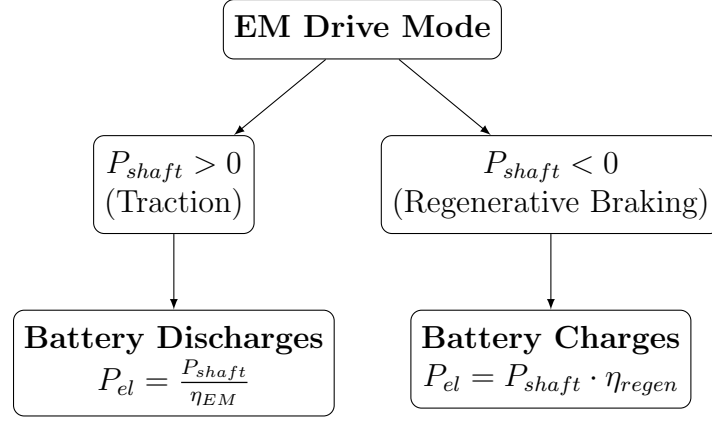


Figure 25: Energy flow logic for Electric Motor (EM) drive.

3.3.2 ICE Drive: Case 1

The ICE delivers exactly the power requested at the wheels at any given moment. The battery does not assist in traction; it is only involved during braking phases to recover energy. Consequently, the engine operates across a wide range of RPM and torque values, often falling into regions of varying efficiency on the engine map.

Whenever the vehicle is slowing down ($P_{shaft} \leq 0$), the ICE is not switched off: the throttle pedal is fully raised, however the Engine still runs at its minimum rpm and consumes the minimum sufficient ($\dot{m}_{min} = 0.12 \cdot 10^{-3} \text{ kg/s}$) amount of fuel to keep it "alive".

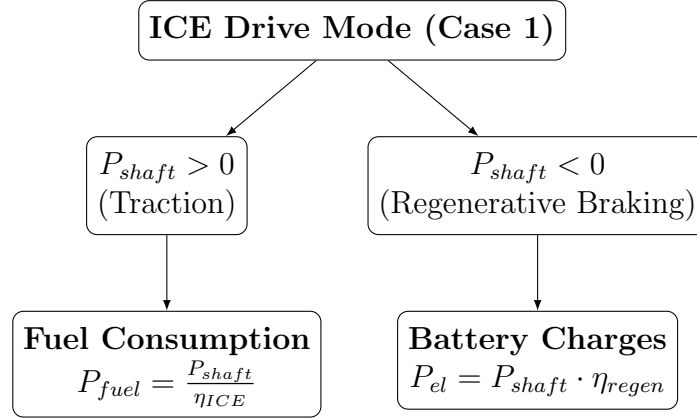


Figure 26: Energy flow logic for ICE Drive Case 1.

The simulation employs an automatic gear selection algorithm designed to optimize fuel efficiency while ensuring vehicle drivability. The logic, implemented in MATLAB, follows a "lowest-RPM first" approach, which prioritizes higher gears to reduce specific fuel consumption:

```

1 %% Gear selection criteria
2 V_ICE = Vprofile(time_ICE + 1);
3 ngear_correct = zeros(length(V_ICE),1);
4 for k = 1:length(ngear_correct)
5     rpm_correct = (60.*V_ICE(k).*gears.*gear_finalf)./(2.*pi.*r);
6     rpm_correct = rpm_correct(rpm_correct >= 1000 & rpm_correct <= 6500)
7 ;

```

```

7   if isempty(rpm_correct)
8       rpm_correct2 = 0;
9       Lspec_correct = 0;
10  else
11      rpm_correct2 = rpm_correct(end);
12      Lspec_correct = (Pshaft(k + length(time_EM))*2*60)/(rpm_correct2
13  *C);
14  end
15  a = 0;
16  while Lspec_correct > min([Lspec(rpm_ICE == round(rpm_correct2)),
17  Lspec(rpm_ICE == (round(rpm_correct2) + 1))])
18      a = a + 1;
19      rpm_correct2 = rpm_correct(end - a);
20      Lspec_correct = (Pshaft(k + length(time_EM))*2*60)/(rpm_correct2*C);
21  end
22  ngear_correct(k) = (2*pi*rpm_correct2*r)/(60*V_ICE(k)*gear_finalf);
23  if isnan(ngear_correct(k))
24      ngear_correct(k) = 0;
25  end
26 end

```

The algorithm operates through the following steps for each time interval:

- **Rpm Feasibility:** for each instantaneous vehicle speed (V), the engine rpm is calculated for all 6 available gear ratios. The algorithm then choose only the gears that keep the engine within its operational limits ($1000 \text{ rpm} \leq n \leq 6500 \text{ rpm}$)
- **Efficiency Optimization:** from the subset of valid gears, the algorithm initially selects the gear corresponding to the lowest engine speed (highest gear ratio). This is based on the principle that lower rpms generally reduce friction losses and improve specific consumption for a given load.
- **Torque Verification and Downshifting:** the system verifies if the engine can sustain the load in the selected gear. It calculates the required specific work (L_{spec}) and compares it against the maximum obtainable specific work at that rpm. If the required load exceeds the engine's maximum capability ($L_{req} > L_{max}$), the algorithm iteratively "downshifts" until a gear is found where the engine allows for the required torque delivery.

The following figure shows the gear shifts adopted at each time instant during the ICE mode:

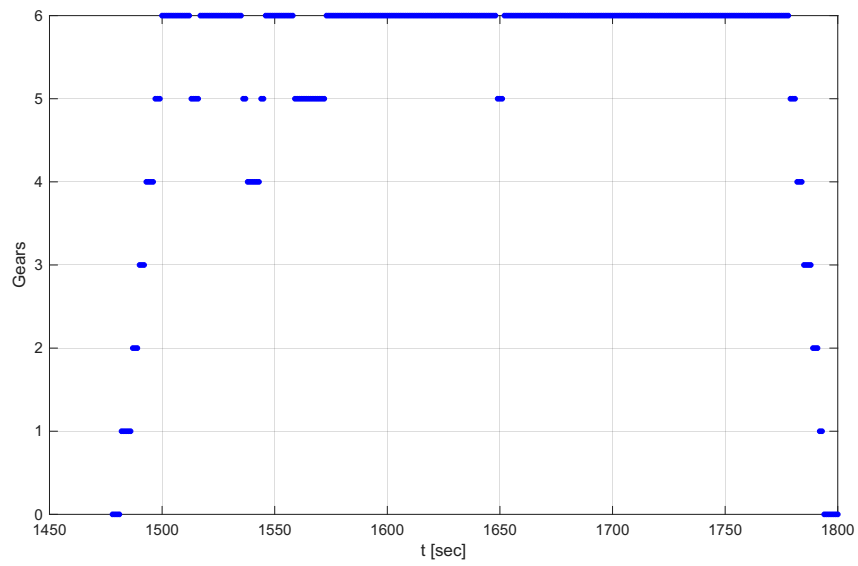


Figure 27: Gear Shifts

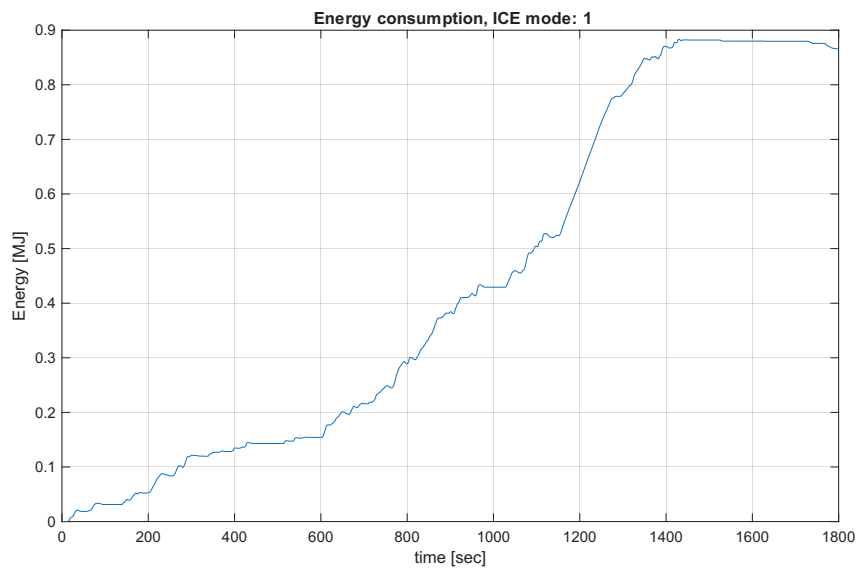


Figure 28: Energy Consumption

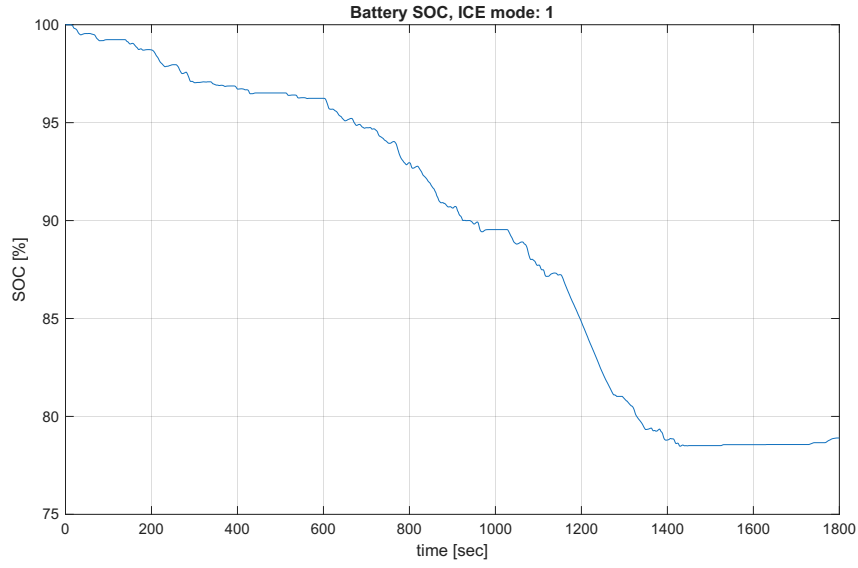


Figure 29: SOC

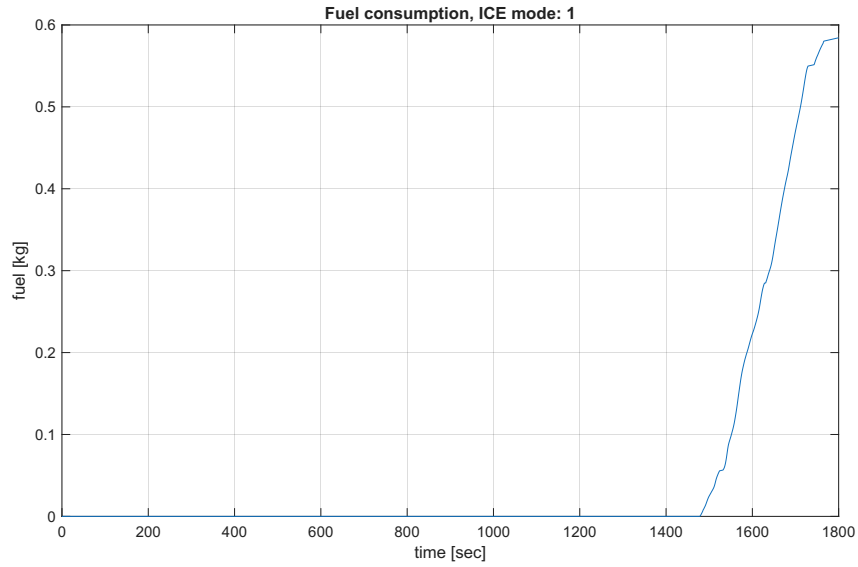


Figure 30: Fuel Consumption

3.3.3 ICE Drive: Case 2

Instead of following the fluctuating road demand, the ICE is set to operate continuously at a fixed, high-efficiency working point. Whenever the engine power is insufficient to meet the traction request, the Electric Motor activates to cover the remaining part. In situations where the engine delivers more power than required, the excess mechanical energy is not wasted but converted by the EM to recharge the batteries.

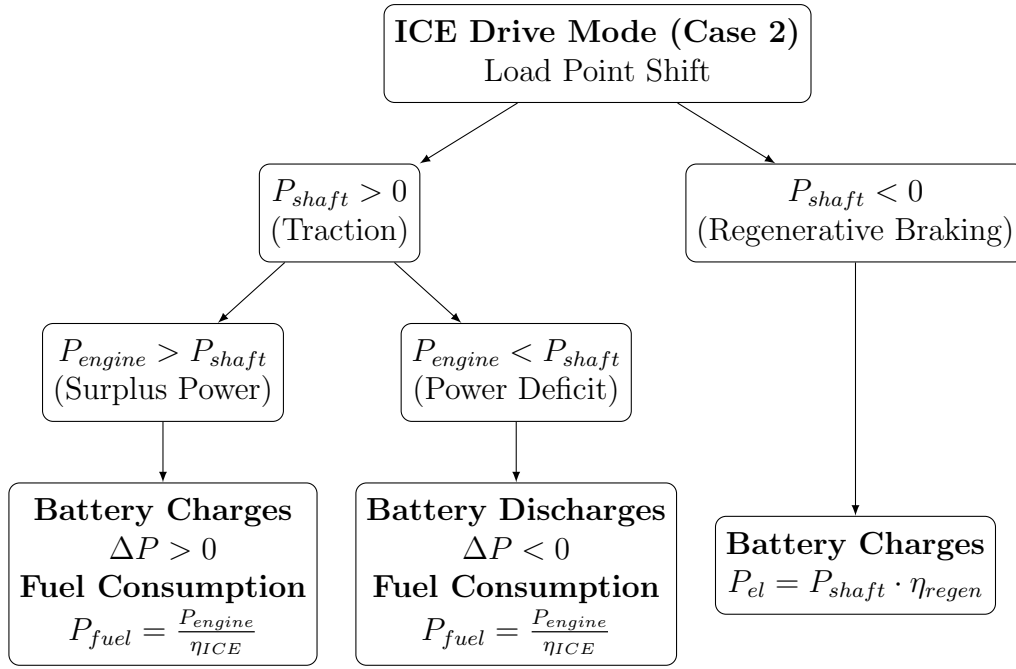


Figure 31: Energy flow logic for ICE Drive Case 2 (Load Shift).

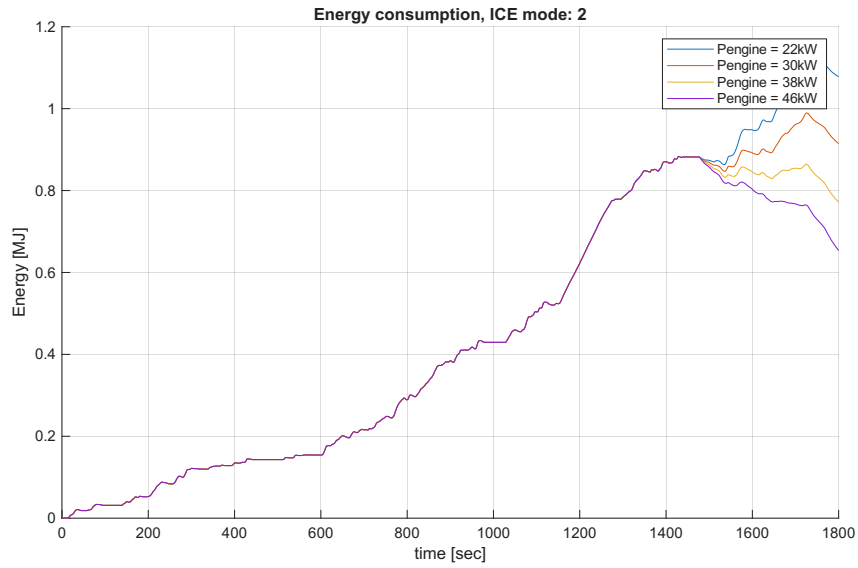


Figure 32: Energy Consumption

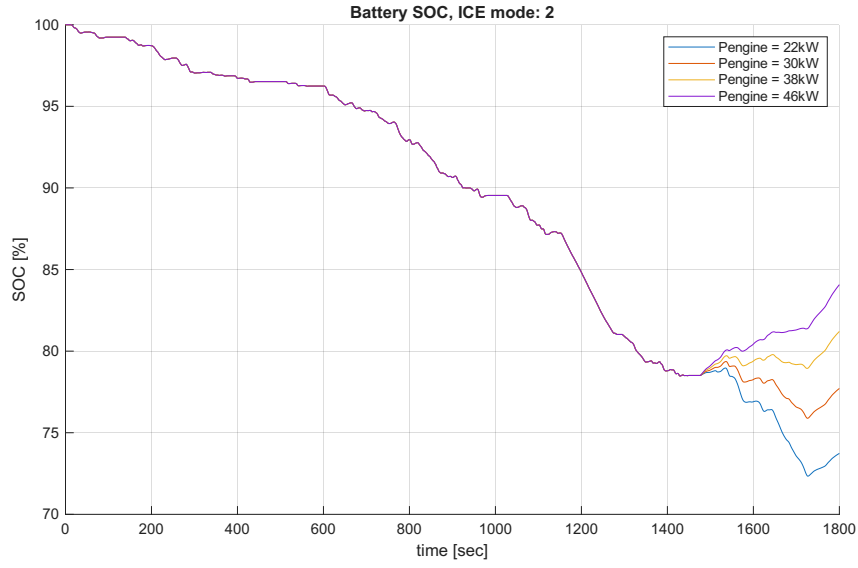


Figure 33: SOC

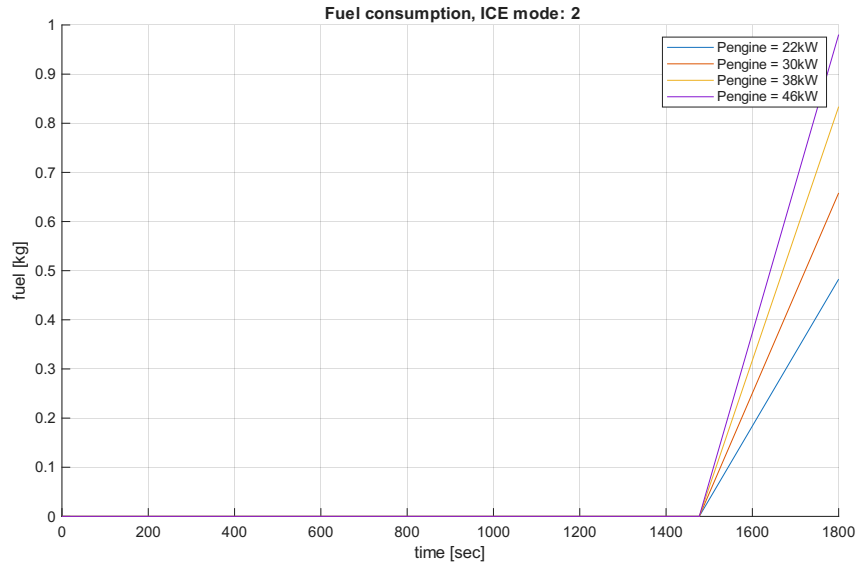


Figure 34: Fuel Consumption

Results and Comparisons

To evaluate the optimal energy management strategy, Case 1 (baseline hybrid operation) was compared against Case 2, where the Internal Combustion Engine (ICE) operates at constant power and optimal efficiency, using the Electric Motor (EM) to balance the required traction power. Performance metrics across different constant power levels (22, 30, 38 and 46 kW) are summarized in Table 4CR1.

Table 4 Performance Comparison: Baseline vs. Constant Power Strategies

Mode / Power	Fuel [kg]	Final SOC [-]	Total Energy [MJ]
Baseline (Case 1)	0.584	0.789	33.66
Case 2 - 22 kW	0.483 (-17.41%)	0.737 (-6.54%)	31.68 (-5.89%)
Case 2 - 30 kW	0.658 (+12.63%)	0.777 (-1.50%)	37.65 (+11.83%)
Case 2 - 38 kW	0.834 (+42.66%)	0.812 (+2.93%)	43.81 (+30.15%)
Case 2 - 46 kW	0.980 (+67.76%)	0.841 (+6.56%)	48.99 (+45.52%)

The analysis of the total energy consumed, calculated by integrating both the fuel and battery power over time, reveals a crucial trade-off between fuel savings and battery charge sustainability.

The 22 kW operating point emerges as the most efficient strategy among the tested scenarios. At this power level, the system achieves a 5.89% reduction in total energy consumption and a significant 17.41% decrease in fuel usage compared to the baseline. However, this relies heavily on battery depletion, as evidenced by a 6.54% drop in the final State of Charge (SOC) relative to Case 1.

Otherwise, as the constant engine power increases (30 kW, 38 kW, and 46 kW), overall efficiency heavily degrades. While the battery SOC increases (up to +6.56% for the 46 kW case), indicating a charge-increasing behavior, the total energy consumption surges by up to 45.52%. This efficiency loss is primarily driven by the "double conversion" penalty. When the ICE generates excess power, that energy must flow through the generator, into the battery, and eventually to the EM for traction. Each component in this chain introduces efficiency losses, compounding the total penalty according to the relation $\eta_{chain} = \eta_{ICE} \cdot \eta_{gen} \cdot \eta_{batt} \cdot \eta_{EM}$.

In conclusion, generating excess electricity onboard via the ICE is energetically expensive and counterproductive for the overall vehicle efficiency. The 22 kW setup represents the optimal balance for this WLTP cycle, effectively minimizing conversion losses while providing sufficient combined thermal and electric power to meet traction demands.

4 Fatigue test on aluminum wheels

4.1 Introduction

The design of automotive wheels requires a sophisticated balance between high-performance metrics: safety, weight optimization, and aesthetics. Among these, safety is the critical priority, defined by superior mechanical strength, fatigue endurance, and impact resistance.

To comply with rigorous international standards, wheels undergo extensive validation protocols. This laboratory focuses on dynamic cornering fatigue testing procedure that combines physical testing machine with Digital Twin simulations to ensure structural integrity.

4.2 Case of study

The test was carried out on an aluminum wheel of a **Porsche 911 Targa 4S** (year 2025).



Figure 35: Porsche 911 Targa 4S

The tested wheel was selected according to the reference tyre **245/35 ZR20 95Y**, usually mounted on this car. The test is performed to analyze the behavior of the wheel during cornering maneuvers, using a dedicated testing machine.



Figure 36: Testing machine of dynamic cornering test

Since this kind of test is time-consuming and expensive, a Digital Twin is realized to fulfill the homologation process reducing the number of physical testing cycle, thus optimizing time and cost. A Digital Twin consists of an accurate numerical model of the real part attached to the test machine based on finite elements approach.

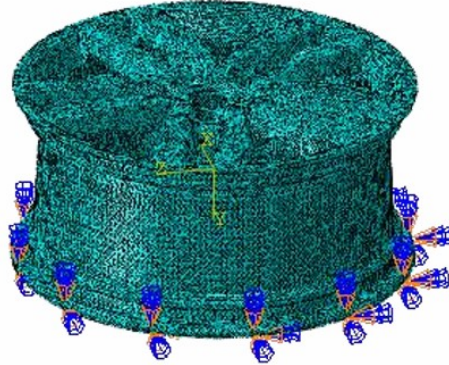


Figure 37: Digital Twin of the wheel

4.3 Testing procedure

To better understand how the process is carried out, the cross section of the machine is provided below.

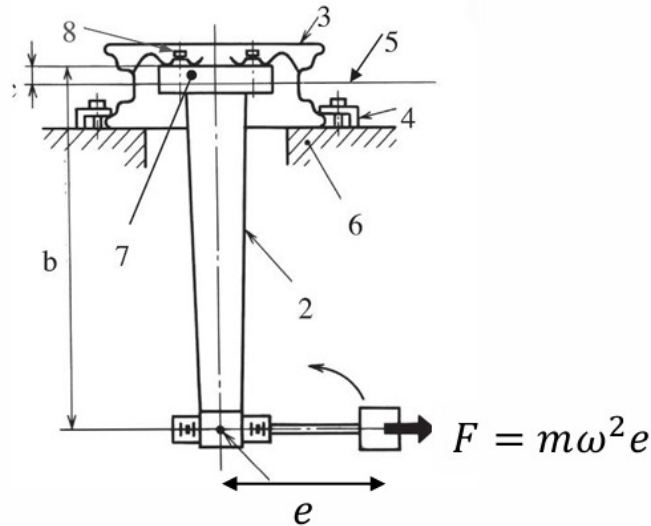


Figure 38: Cross section of testing machine

As shown, the bottom part of the wheel rim is clamped on a plane. The wheel is then coupled via bolts to a solid loading axle, which is integrated with a rotating mass system. This configuration induces a rotating bending moment—and consequently, cyclic fatigue stress—that is transmitted through the shaft directly into the wheel structure.

4.3.1 Bending moment computation

The total bending moment is determined by the superposition of two primary loading components: the vertical load acting on the tyre contact patch and the lateral friction force generated during cornering.

The analytical expression used for the assessment is as follows:

$$M = (\mu \cdot R + d) \cdot F_v \cdot S \quad (4)$$

- M : bending moment [Nm];
- μ : friction coefficient between tyre and road [-];
- R : tyre radius [m];
- d : inset or outset of the wheel (positive for inset, negative for outset) [m];
- F_v : maximum vertical static load acting on the wheel [N];
- S : accelerated test factor [-].

The tyre radius R can be obtained by the data represented in the tyre code. It is obtained by summing the radius of the wheel rim and the sidewall height. The final value of the tyre radius is the following:

- rim radius: 10 in = 254 mm;
- tyre width: 245 mm;
- sidewall height: 35% of the tyre width = 85.75 mm;

$$R = \text{rim radius} + \text{sidewall height} = 339.75 \text{ mm}$$

The inset d is taken from the wheel rim code, which is **8.5Jx20 ET50**. So, d is equal to **50 mm**.

The maximum vertical load F_v is given by the load index of the tyre. In this case, the load index is 95 that corresponds to a maximum load of **690 kg**.

The accelerated test factor S is a safety coefficient that depends on the material chosen for manufacturing the wheel. According to the standards, the factor must be selected based on the minimum number of cycles to which the wheel must resist. For an aluminum rim required to withstand a minimum of 50,000 cycles, S is set to **2**.

The following table summarizes the parameters utilized for the final bending moment calculation with the correct unit of measurement.

Parameter	Value
Friction coefficient μ	0.7
Tyre radius R	0.34 m
Inset of the wheel d	0.05 m
Vertical static load F_v	6769 N
Accelerated test factor S	2
Bending moment M	3897 Nm

Table 5: Parameters for bending moment calculation

4.3.2 Fatigue stress computation

Stress analysis was performed via Finite Element Analysis (FEA) utilizing quadratic tetrahedral elements. The methodology involves computing the dynamic stress tensor as a linear combination of two orthogonal unitary bending moments applied at the wheel center across distinct simulations.

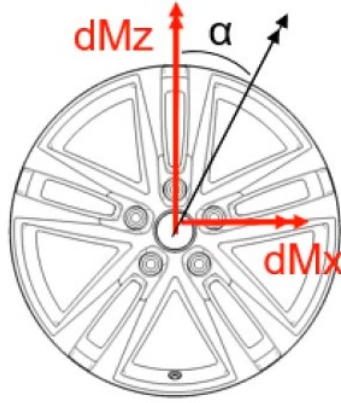


Figure 39: Representation of unitary bending moments

α The variable α represents the angular position ($0 \leq \alpha \leq 2\pi$). The resulting stress state from the unitary simulations is then scaled by the calculated test bending moment M .

The final formula is shown below.

$$\sigma(\alpha) = M(\sigma_{dMz}\cos\alpha + \sigma_{dMx}\sin\alpha) \quad (5)$$

- $\sigma(\alpha)$ = component of dynamic stress tensor [MPa];
- M = bending moment [Nm];
- $\sigma_{dMz}, \sigma_{dMx}$ = stress states from unitary simulations [MPa/Nm];
- α = angular position [rad].

The stress tensor was computed in MATLAB for an angular discretization of 1000 steps. Unitary stress components were retrieved from the WheelStressAndMaterialData.mat

dataset, which contains nodal coordinates and the six independent components of the stress tensor $(\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{xz}, \tau_{yz})$.

The multiaxial fatigue state was then evaluated using the Sines criterion:

$$\sigma_S = \sqrt{\sigma_{x,a}^2 + \sigma_{y,a}^2 + \sigma_{z,a}^2 - (\sigma_{x,a}\sigma_{y,a} + \sigma_{x,a}\sigma_{z,a} + \sigma_{y,a}\sigma_{z,a}) + 3(\tau_{xy}^2 + \tau_{xz}^2 + \tau_{yz}^2)} \quad (6)$$

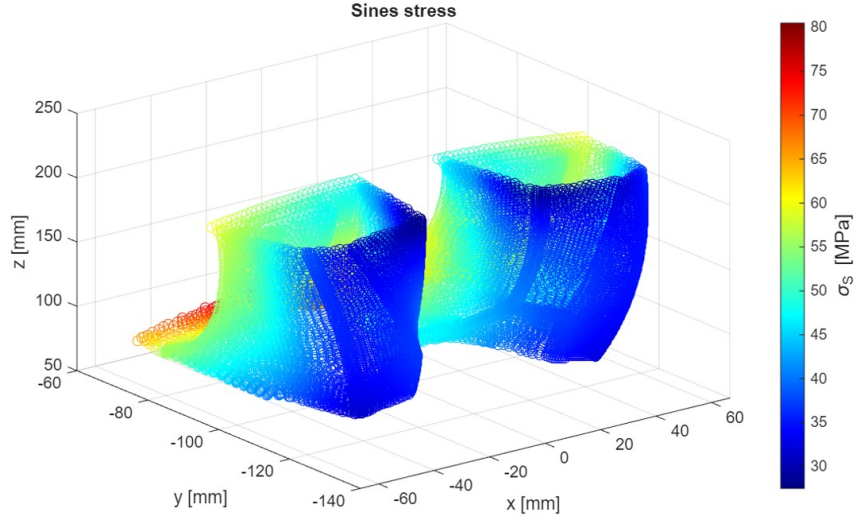
- σ_S = equivalent Sines fatigue stress [MPa];
- $\sigma_{x,a}, \sigma_{y,a}, \sigma_{z,a}$ = alternating principal stresses [MPa];
- $\tau_{xy}, \tau_{xz}, \tau_{yz}$ = alternating shear stresses [MPa].

The alternating stress components were derived as:

$$\sigma_a = \frac{\sigma_{max} - \sigma_{min}}{2} \quad (7)$$

- σ_a = alternating stress [MPa];
- σ_{max} = maximum stress value [MPa];
- σ_{min} = minimum stress value [MPa].

Numerical results are visualized through 3D scatter plots, highlighting the cyclic symmetry of the stress distribution across the wheel geometry.



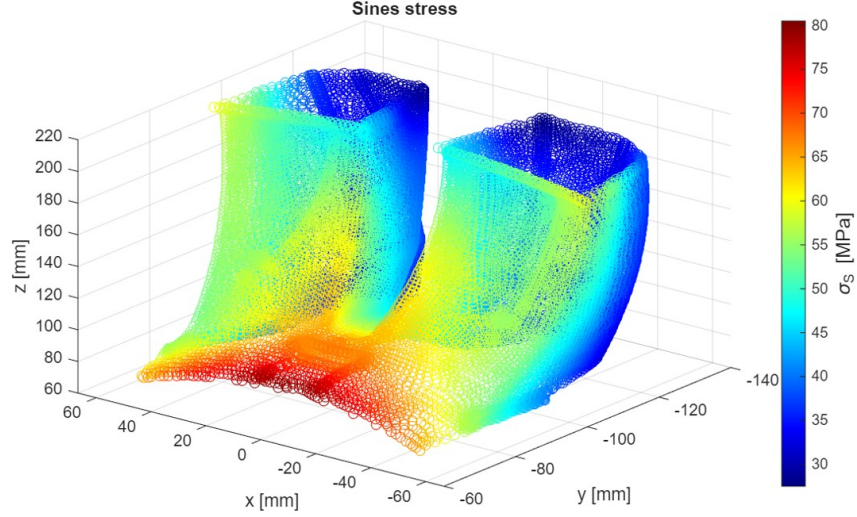


Figure 40: Front and rear plots of fatigue state of stress

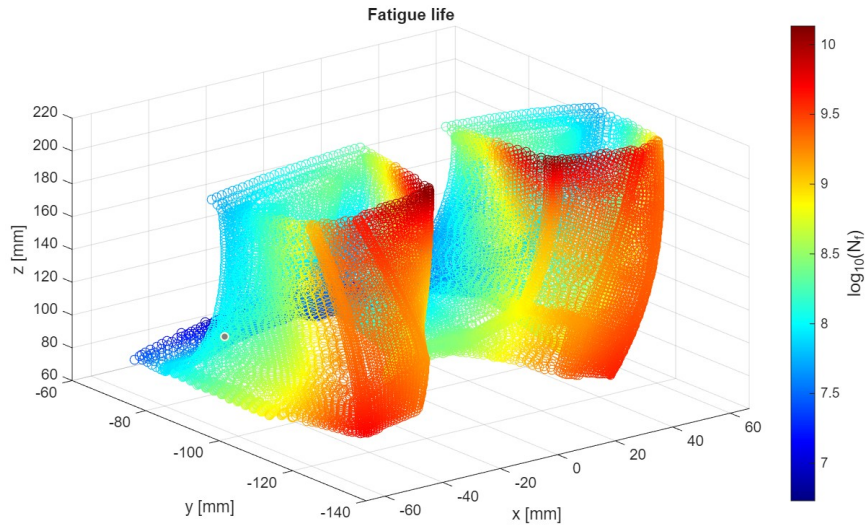
4.3.3 Fatigue life computation

The operational life of the wheel was estimated using the Wöhler curve (S-N curve) relationship:

$$N_f = \frac{1}{2} \left(\frac{\sigma_S}{\sigma'_f} \right)^{\frac{1}{b}} \quad (8)$$

- N_f = number of fatigue cycles;
- σ_S = equivalent Sines stress [MPa];
- σ'_f = fatigue limit strength [MPa];
- b = slope of the Wohler's curve.

The spatial distribution of the fatigue life is mapped onto the wheel geometry as shown below.



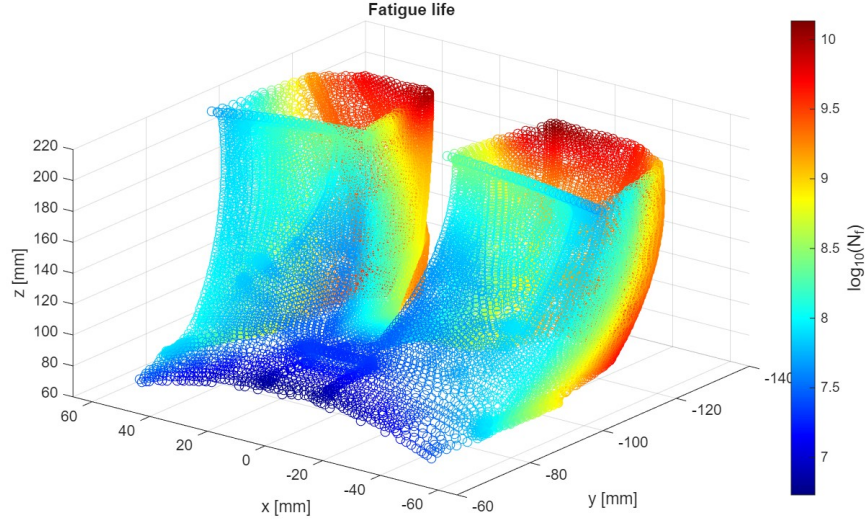


Figure 41: Front and rear plots of fatigue life of the wheel

4.4 Conclusion and final comments

The assessment confirms an inverse correlation between the equivalent stress state and the predicted fatigue life.

Critical stress concentrations were identified primarily within the fillet region at the junction between the hub mounting face and the internal spoke profiles. This region represents the most vulnerable site for potential fatigue crack initiation.

The table below summarizes the coordinates and fatigue metrics for the most critical node identified in the simulation.

Parameter	Value
x coordinate	0.4995 mm
y coordinate	-65.1231 mm
z coordinate	87.7605 mm
Sines fatigue stress	80.5 MPa
$\log_{10}N_f$	6.73

Table 6: Fatigue assessment of most critical point

5 Braking system laboratory

5.1 Introduction

This laboratory session focused on the braking dynamics of motorcycles. The primary objective was to evaluate the critical parameters influencing braking performance, specifically regarding **optimal braking force distribution** and the **design of core components** including the brake lever, calipers, and discs.

5.2 Optimal repartition analysis

Optimal brake force repartition defines the ideal ratio between front and rear braking effort to maximize deceleration.

The calculation of the key parameters is based on the dynamics model presented in the following picture, which shows the main forces acting on the vehicle.

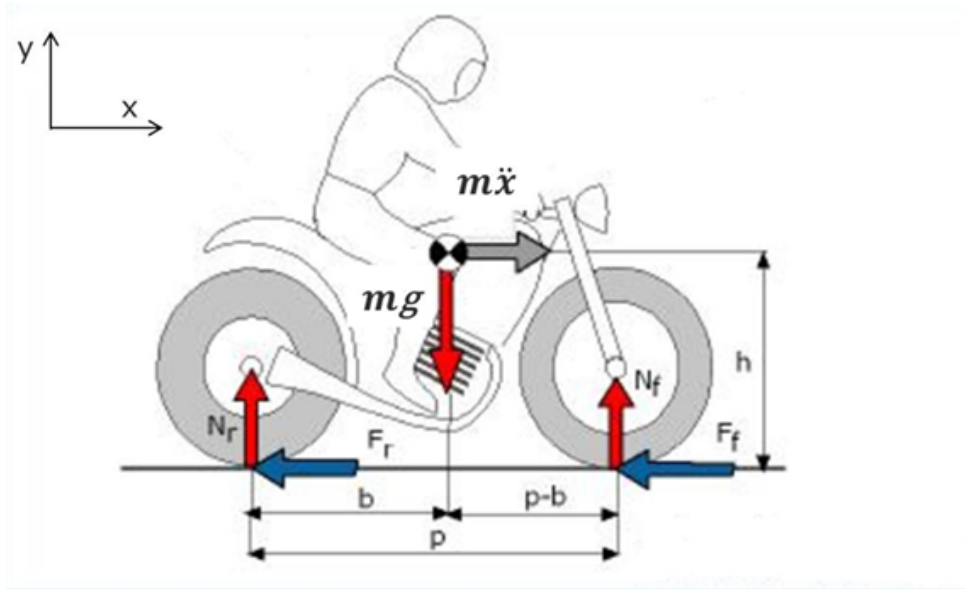


Figure 42: Motorbike dynamics model

- mg : motorbike weight [N];
- $m\ddot{x}$: inertia force [N];
- F_f, F_r : front and rear braking forces [N];
- N_f, N_r : front and rear vertical forces [N];
- p : wheelbase [m];
- $p - b$: distance of the center of gravity from the front axle [m];
- b : distance of the center of gravity from the rear axle [m];
- h : height of the center of gravity [m].

The main parameters of the analysis are reported in the following table.

Parameter	Value
Wheelbase p	1435 mm
CoG distance from rear axle b	665 mm
CoG height h	677 mm
Bike Mass m_{bike}	157 kg
Rider Mass m_{rider}	70 kg
Total Mass m	227 kg
Road Friction Coefficient μ	0.7
Maximum velocity v	315 km/h
Drag coefficient C_D	0.6
Frontal area A	0.4 m ²
Air density ρ	1.2 kg/m ³

Table 7: Vehicle and Rider Technical Specifications

These terms were used to determine essential parameters, listed below, for assessment of the braking repartition.

- **Static load distribution** $\frac{\Delta F_{z,st}^{front}}{\Delta F_{z,st}^{rear}}$

$$\Delta F_{z,st}^{front} = \frac{b}{p} \quad (9)$$

$$\Delta F_{z,st}^{rear} = \frac{p-b}{p} \quad (10)$$

Result: **46/54**

- **Deceleration at flipping limit** $\ddot{x}_{flip,max}(g)$, without considering aerodynamic effects

$$\ddot{x}_{flip,max}(g) = \frac{p-b}{h} \quad (11)$$

Result: **1.14g**

Flipping is the condition in which the rear wheel is no longer in contact with the ground.

- **Maximum deceleration** $x_{max}(g)$

$$\ddot{x}_{max}(g) = \mu \quad (12)$$

Result: **0.7**

- **Brake repartition needed to reach the maximum deceleration** $\frac{F_f}{F_r}$

$$\frac{F_f}{mg} = \frac{b + \mu h}{p} \quad (13)$$

$$\frac{F_r}{mg} = \frac{(p-b) - \mu h}{p} \quad (14)$$

Result: **79/21**

Further information about the braking repartition is deductible from the following graph, which relates the front and rear braking distribution with the adimensional total braking forces, i.e. maximum and flipping decelerations.

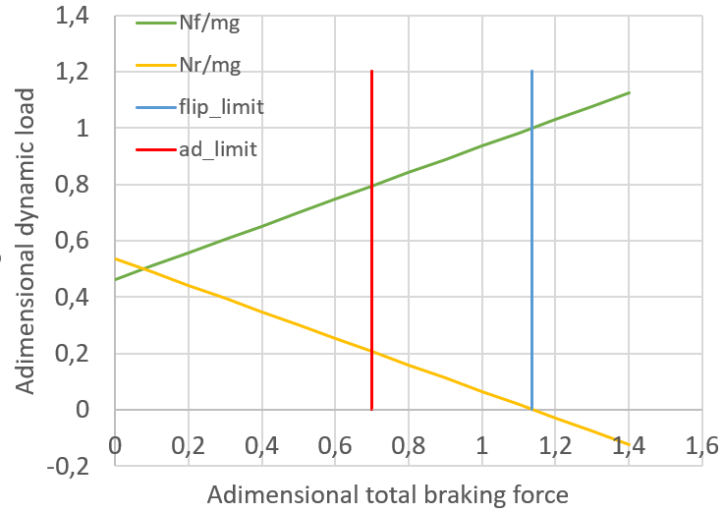


Figure 43: Braking distribution

- Maximum deceleration reached by using only the front brake $\ddot{x}_{max}^{front}(g)$

$$\ddot{x}_{max}^{front}(g) = \frac{\mu b}{p - \mu h} \quad (15)$$

Result: **0.48g**

If only front brake was used, the maximum deceleration curve would shift as showed in the picture below.

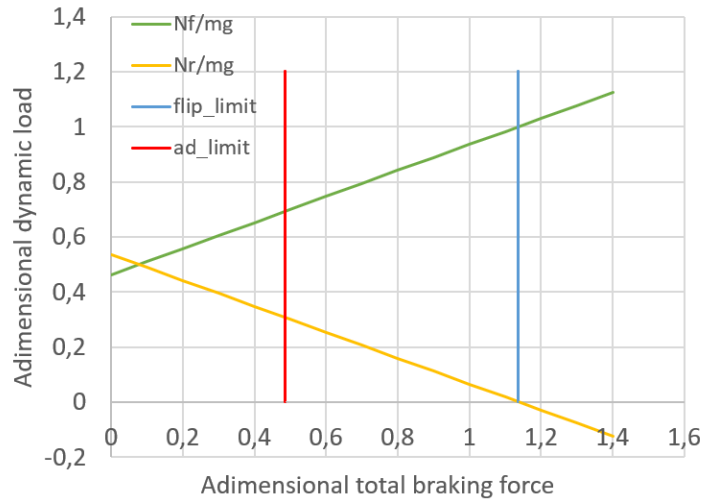


Figure 44: Braking distribution for only front brake

- Maximum deceleration reached by using only the rear brake $\ddot{x}_{max}^{rear}(g)$

$$\ddot{x}_{max}^{rear}(g) = \frac{\mu(p - b)}{p + \mu h} \quad (16)$$

Result: **0.28g** Even in this case, the maximum deceleration curve is shifted.

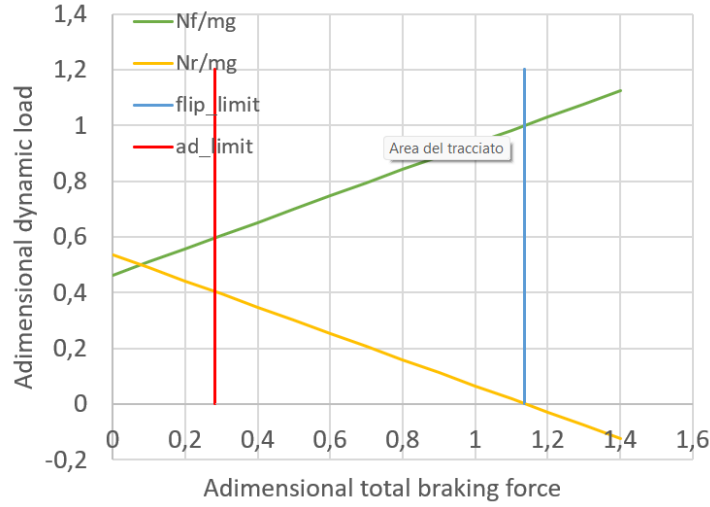


Figure 45: Braking distribution for only rear brake

- Deceleration at flipping limit considering aerodynamic effects $\ddot{x}_{flip,max}^{aero}(g)$

$$\ddot{x}_{flip,max}^{aero}(g) = \frac{p-b}{h} + \frac{F_D}{mg} \quad (17)$$

$$F_D = \frac{1}{2}\rho C_D A v^2 \quad (18)$$

Result: **1.63g**

The drag force helps the motorcycle stop, thus **increasing the limit deceleration and stability at high speeds**. So it is possible to brake more before flipping.

The final braking point is depicted in the plot below, considering the ideal curve as well.

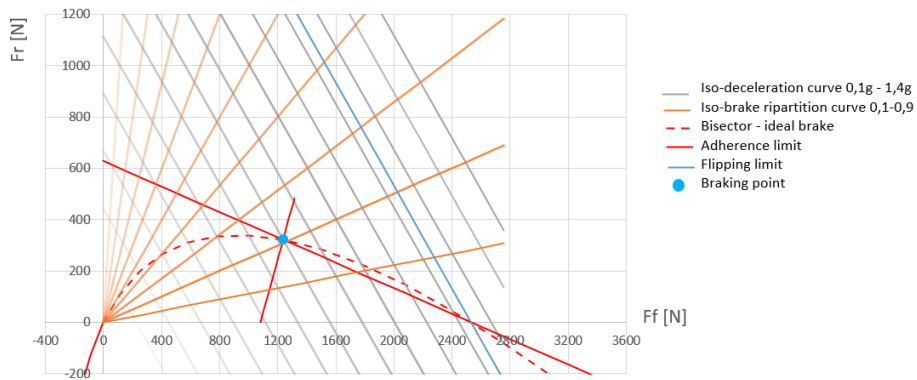


Figure 46: Braking point and ideal braking curve distribution

5.3 Braking system design

The braking system is composed by two discs and calipers with the following characteristics:

- **Disc diameter** D_{disc} : 340 mm;
- **Calipers diameters** D_1, D_2 : 36/38 mm;
- **Disc/pad friction coefficient** $\mu_{disc-pad}$: 0.5;
- **Deceleration at flipping limit** $\ddot{x}_{flip,max}$: 16 m/s²;
- **Wheel radius** R_{tyre} : 300 mm;

First of all, the system pressure at flipping limit was computed considering the braking longitudinal force achievable F_x .

$$F_x = m\ddot{x}_{flip,max} \quad (19)$$

Then, the braking torque applied on the wheel T_b can be used to relate the longitudinal braking force F_x with the tangential force on the pads F_{tan} .

$$T_b = F_x R_{tyre} = 2F_{tan} R_{eff} \quad (20)$$

R_{eff} is the **effective radius of the disc** equal to half the disc diameter D_{disc} .

From the force F_{tan} it is possible to obtain the force applied by the pistons F_{normal} , function of the pressure p and the piston diameters D_1 and D_2 .

$$F_{normal} = \frac{F_{tan}}{\mu_{disc-pad}} \quad (21)$$

$$p = \frac{F_{normal}}{2\frac{\pi}{4}(D_1^2 + D_2^2)} \quad (22)$$

The final pressure p is equal to **11.4 bar**.

Then, the rider force applied at brake lever F_{rider} was calculated considering two different master cylinders (M/C).

The following figure represents a scheme explaining how the force is applied to the brake lever.

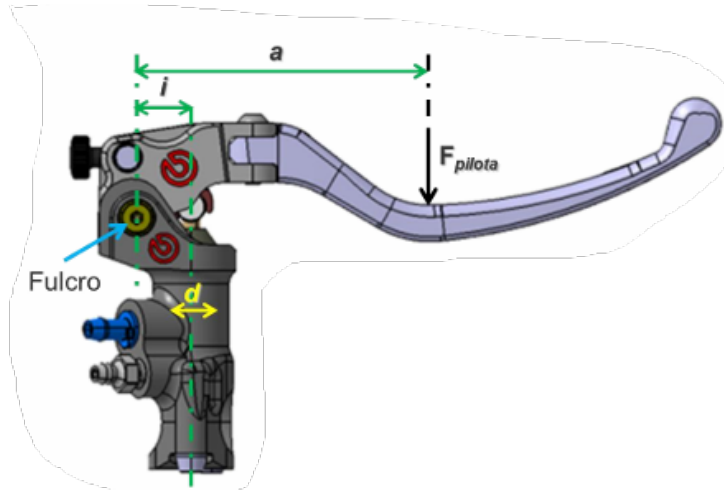


Figure 47: Braking force on the lever

The final force is computed assuming that:

- the distance between the rider's finger from the M/C fulcrum a is 100 mm;
- the M/C internal spring is preloaded;
- inefficiencies due to friction of the internal components increase the force value by 10%.

The nominal force F_{rider} can be obtained according to the formula below.

$$F_{rider} = p \frac{\pi}{4} d^2 \frac{i}{a} \quad (23)$$

d is the **M/C piston diameter** in mm and i is the **M/C pivot distance** in mm.

The M/Cs characteristics are reported in table 7, including the total force on the lever.

Parameter	19x18 M/C	19x20 M/C
M/C piston diameter d	19 mm	19 mm
M/C pivot distance i	18 mm	20 mm
Rider force F_{rider}	68.6 N	76.2 N

Table 8: Master pistons characteristics

According to the results, the 19x18 master piston enables the flipping limit to be attained by applying less force on the brake lever. This offers many advantages, such as **reduced rider fatigue** during long races and **improved modulation of the force** near the limit.

6 McPherson suspension assignment

6.1 Introduction

This assignment regards the design of a McPherson suspension for a front drive wheel car. The objective is to realize the assembly drawing and the production drawing of the lower arm.

The final design is a whole combination of phases, starting from the positioning of the suspension joints, then moving to the estimation of the forces acting on the system and the structural verification of the lower arm.

6.2 Positioning of kinematic points

6.2.1 Definition of the task

The first part of the assignment focuses on the **placement of the kinematic points** of the suspension, where the main joints have been located.

The correct disposition was performed by taking into account a dynamic study on the vehicle, in which some requirements on angles as function of the wheel travel have been defined. The objective was to converge on the **target angle values**, dealing with constraints on the allowed locations of the points.

The required target angles are exposed in the following table.

	Nominal position	Full compression (80 mm)	Full extension (95 mm)
Camber angle	-1°	-0°26'30"	+0°28'24"
Steering angle	-0°03'17"	-0°46'22"	+0°26'10"

Table 9: Kinematic objectives

The coordinates of the points were determined according to a reference system, which has origin in a point belonging to the mean plane of the vehicle, x axis along the longitudinal direction of the vehicle rearwards, z axis vertical upward and y axis as to create a right-handed system.

Table 7 shows all the coordinates of the kinematic points and the range of variation of the design variables of points to be placed, which are:

- coordinates y and z of point B;
- coordinates y and z of point C1;
- coordinates y of point G.

The coordinates of point C2 are function of the ones of C1, as reported in the table.

Point	Description	Coordinates (x,y,z) [mm]
A	Rotation centre of the upper bushing	(31.8, 593, 604)
B	Joint between wheel hub carrier and lower arm	(-9.10, 715 ÷ 735, -50 ÷ -70)
C1	Joint between chassis and lower arm	(-10, 390 ÷ 410, -45 ÷ -65)
C2	Joint between chassis and lower arm	(311, y_{C1} , $z_{C1} + 4.6$)
D	Tyre-road contact point	(-1.6, 773.5, -226.8)
E	Wheel centre	(-1.6, 768.4, 65.1)
F	Steering joint on the wheel hub carrier	(136.9, 682.7, 88.5)
G	Steering joint on the chassis (centred steer)	(170, 300 ÷ 320, 86)
O	Reference system origin	(0, 0, 0)

Table 10: Hardpoints coordinates and descriptions

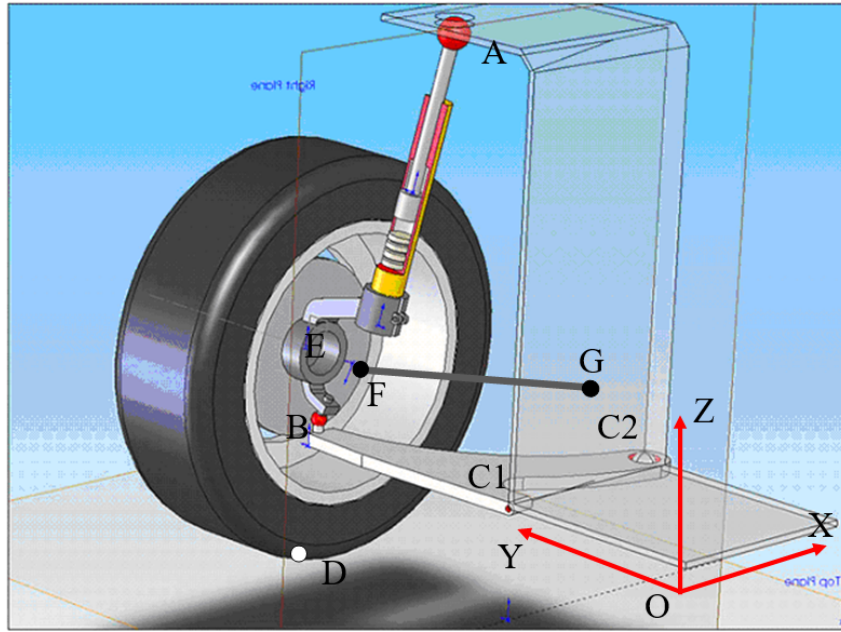


Figure 48: Hardpoints location layout

The inclination of the damper is $3^{\circ}32'30''$ forward and $2^{\circ}30'48''$ outward in the nominal configuration of the suspension. The reference tyre size is 205/60R15.

6.2.2 Computation algorithm and final results

To position correctly the kinematic points, a MATLAB algorithm was implemented.

It consists of an approximation model based on **Artificial Neural Networks** (ANNs), called `mcpersonnn_fn`, that takes the hardpoints coordinates as input and gives the suspension steer and camber curves as output.

Since the objective was to get as close as possible to the target values, the ANN model was embedded in a two-stage optimization scheme, utilizing a **genetic algorithm** (ga) for global exploration followed by a **constrained nonlinear optimization function** for local refinement (`fmincon`). The constraints were given by the range of variations of the points to be determined.

The aim was to minimize as much as possible the **mean square error** (MSE) between the output angles of the ANN model and the objective values, according to the equation below:

$$MSE = \frac{1}{n} \sum_{i=1}^n [(\gamma_{calc,i} - \gamma_{target,i})^2 + (\delta_{calc,i} - \delta_{target,i})^2] \quad (24)$$

- MSE : Total Mean Squared Error (Fitness value to be minimized).
- n : Number of sampling points along the wheel travel ($n = 3$).
- $\gamma_{calc,i}$: Computed **Camber angle** at the i -th point.
- $\gamma_{target,i}$: Target **Camber angle** at the i -th point.
- $\delta_{calc,i}$: Computed **Steering angle** at the i -th point.
- $\delta_{target,i}$: Target **Steering angle** at the i -th point.

A detailed implementation of the aforementioned algorithm is provided in the following section.

```

1 %% McPherson Optimization
2
3 clear
4 clc
5 close all
6
7 %% TARGETS
8 WT_target = [80 0 -95]; % wheel travel
9 Camber_target = [-(26/60+30/3600) -1 (28/60+24/3600)]; % camber target
10 Steer_target = [-(46/60+22/3600) -(3/60+17/3600) (26/60+10/3600)]; %
    steer target
11 save('targets.mat','WT_target','Camber_target','Steer_target');
12
13 %% VARIABLES CONSTRAINTS
14
15 lb = [390 -65 300 715 -70]; % lower bounds
16 ub = [410 -45 320 735 -50]; % upper bounds
17
18 %% OBJECTIVE FUNCTION FOR THE GA
19
20 mse_local = @(x) mse_global(x, WT_target, Camber_target, Steer_target);
21
22 % it finds a of good local minimum
23
24 %% STEP 1 Genetic Algorithm (GA)
25
26 optsGA = optimoptions('ga', ...
27     'Display','iter', ...
28     'UseParallel', true, ...
29     'PopulationSize', 100, ...
30     'MaxGenerations', 150);
31 [x_ga, err_ga] = ga(mse_local, 5, [], [], [], [], lb, ub, [], optsGA);
32
33 %% STEP 2 fmincon (local refinement)
34
35 optsF = optimoptions('fmincon', ...
36     'Display','iter', ...
37     'Algorithm','sqp', ...

```

```

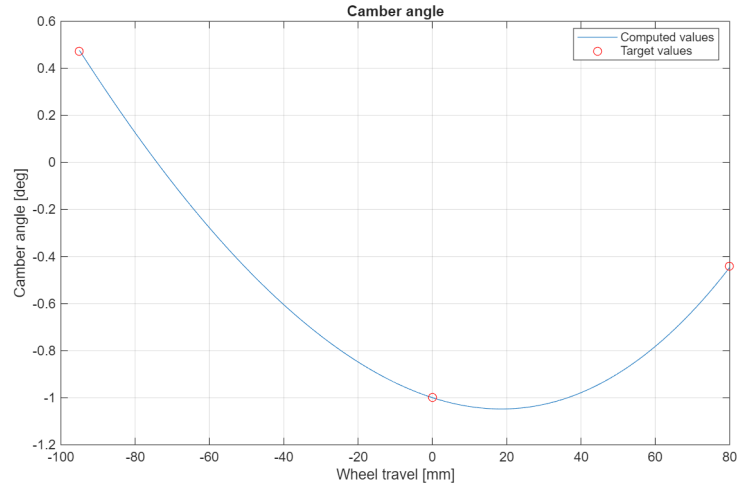
38     'MaxFunctionEvaluations', 3000);
39 [x_final, err_final] = fmincon(mse_local, x_ga, [], [], [], [], lb, ub,
    [], optsF);
40
41 % it finds the global minimum
42
43 %% RESULTS SAVING
44
45 save('mcperson_results.mat', 'x_final', 'err_final', 'x_ga', 'err_ga');
46
47 % the results are stored in matrix mcperson_result.mat including the
    final optimal coordinates, the final mse and the results of the ga
48
49 %% OBTAINED CURVES PLOTS
50
51 y_C1 = x_final(1);
52 z_C1 = x_final(2);
53 y_G = x_final(3);
54 y_B = x_final(4);
55 z_B = x_final(5);
56
57 [WT_out, y_out, Camber_out, Steer_out] = mcpersonNN_fn(y_C1, z_C1, y_G,
    y_B, z_B);
58
59 % Plot WheelTravel vs Camber angle
60
61 figure(1)
62 plot(y_out(:,1), y_out(:,2), WT_target, Camber_target, 'or'), grid
63 title('Camber angle')
64 xlabel('Wheel travel [mm]')
65 ylabel('Camber angle [deg]')
66 legend('Computed values', 'Target values', 'location', 'best')
67
68 % Plot WheelTravel vs Steering angle
69
70 figure(2)
71 plot(y_out(:,1), y_out(:,3), WT_target, Steer_target, 'or'), grid
72 title('Steering angle')
73 xlabel('Wheel travel [mm]')
74 ylabel('Steering angle [deg]')
75 legend('Computed values', 'Target values', 'location', 'best')
76
77 %% LOCAL OBJECTIVE FUNCTION FOR fmincon
78
79 function err = mse_global(x, WT_target, Camber_target, Steer_target)
80
81     y_C1 = x(1);
82     z_C1 = x(2);
83     y_G = x(3);
84     y_B = x(4);
85     z_B = x(5);
86     [WT_out, ~, Camber_out, Steer_out] = mcpersonNN_fn(y_C1, z_C1, y_G,
        y_B, z_B);
87     if ~isequal(numel(WT_out), numel(WT_target))
88         error('The number of points returned by mcpersonNN_fn does not
            match the WT_target size.');
```

```

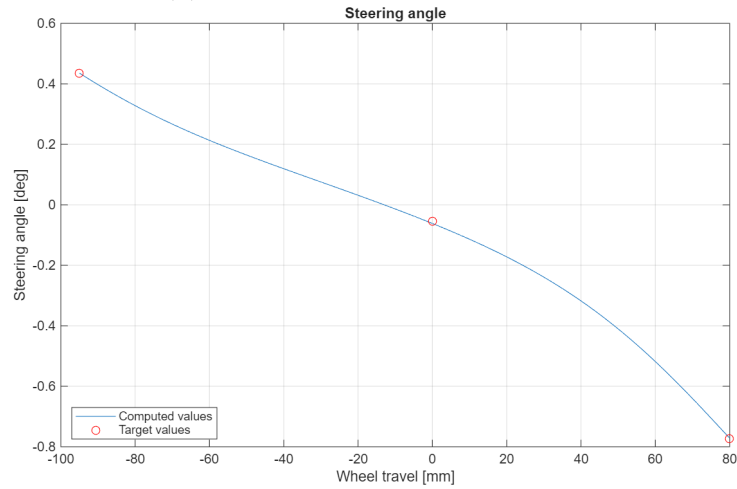
Steer_target).^2 );
91
92 % mse computation
93
94 end

```

The obtained curves and the corresponding optimal coordinates are illustrated in the figures and table below.



(a) Camber angle vs wheel travel



(b) Steer angle vs wheel travel

Figure 49: Final kinematic curves.

Coordinate	Value [mm]
y_{C_1}	407.67
z_{C_1}	-51.87
y_G	319.25
y_B	725
z_B	-58

Table 11: Final coordinates

6.3 Forces Estimation

6.3.1 System Modeling and Simplifying Assumptions

The objective of this analysis is to determine the forces and moments exchanged between the McPherson suspension components under aggravated loading conditions. The system has been modeled with the following assumptions:

- **External Forces:** The forces deriving from the tire-road interaction (Vertical force Z , lateral Y , longitudinal X) are applied solely at the contact patch **D**.
- **Rigid Elements:** The spring-damper assembly is considered as a rigid element integral with the hub carrier along the steering axis passing through **A** and **B**.
- **Joint B:** The connection between the hub carrier and the lower control arm (point **B**) is treated as an ideal spherical joint, capable of transmitting only spatial forces but no moments.

The lower control arm is physically constrained to the chassis at two distinct points (**C1** and **C2**). To make the system analytically solvable, the approach of virtually collapsing the two attachment points into a single **equivalent pole located at C1** was adopted.

This approach transforms the ground constraint of the lower control arm into a boundary condition that must provide both force reactions (X_{c1}, Y_{c1}, Z_{c1}) and moment reactions ($M_{C1x}, M_{C1y}, M_{C1z}$). The 13-unknown vector of the system ($\mathbf{x}$) is therefore defined as:

$$\mathbf{x} = [X_a \ Y_a \ Z_a \ X_b \ Y_b \ Z_b \ X_{c1} \ Y_{c1} \ Z_{c1} \ M_A \ M_{C1x} \ M_{C1y} \ M_{C1z}]^T \quad (25)$$

The structure of the system reflects the mechanical decoupling between the upright-damper assembly and the lower control arm.

The upper mount **A** reacts with an unknown steering torque M_A to prevent free rotation of the hub carrier around the steering axis. In the mathematical matrix, this moment has been decomposed into its three spatial components by multiplying the unknown magnitude by the unit vector of the steering axis (calculated using the camber α and caster β angles).

The directional unit vector of the steering axis, $\mathbf{v}_{out}$, is defined using a normalization factor k :

$$k = \sqrt{\tan^2(\beta) + \tan^2(\alpha) + 1} \quad (26)$$

$$\mathbf{v}_{out} = \frac{1}{k} \begin{bmatrix} \tan(\beta) \\ -\tan(\alpha) \\ 1 \end{bmatrix} \quad (27)$$

The unknown scalar torque M_A is then multiplied by this unit vector to obtain the three spatial moment components acting on the hub carrier:

$$M_{Ax} = M_A \cdot \frac{\tan(\beta)}{k} \quad (28)$$

$$M_{Ay} = M_A \cdot \frac{-\tan(\alpha)}{k} \quad (29)$$

$$M_{Az} = M_A \cdot \frac{1}{k} \quad (30)$$

These three components are substituted directly into the global rotational equilibrium equations (Rows 4, 5, and 6 of the system matrix).

Similarly, since point **B** is modeled as a spherical joint, the lower control arm is structurally isolated from the moments applied to the hub carrier. The equivalent constraint at **C1** reacts exclusively to the bending moment generated by the spatial forces applied at **B**.

6.3.2 Matrix Formulation

The global equilibrium of the assembly (rows 1-6), the equilibrium of the isolated lower control arm (rows 7-12), and the kinematic condition of free rotation around the C1-C2 axis (row 13) Since pole **C1** is a “fictitious” constraint that also encompasses **C2**, it is necessary to impose a fundamental final kinematic condition: the lower control arm is a hinged structure that must be free to rotate to accommodate the suspension wheel travel.

Therefore, it is enforced that **the resultant reaction moment at C1, projected along the axis connecting C1 and C2, must be strictly zero**. Mathematically, this is expressed by setting the dot product (scalar product) between the moment vector at **C1** and the direction vector of the **C2-C1** axis to zero:

$$\mathbf{M}_{C1} \cdot \mathbf{v}_{C2C1} = 0 \quad (31)$$

Expanding this dot product into its spatial components yields the 13th and final equation of the linear system:

$$\boxed{M_{C1x} \cdot v_{C2C1,x} + M_{C1y} \cdot v_{C2C1,y} + M_{C1z} \cdot v_{C2C1,z} = 0} \quad (32)$$

This mathematical condition is crucial: it effectively closes the analytical problem by allowing the linear system to “understand” the rotational degree of freedom, correctly partitioning the constraints while ensuring the control arm behaves as a mechanism free to pivot around its physical chassis mounts. These equations generate a linear system in the form $\mathbf{A} \cdot \mathbf{x} = \mathbf{b}$.

The coefficient matrix **A** (constructed using the relative geometric distances between the points, e.g. $d_{AC1,z} = |A_z - C1_z|$) is structured as follows:

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -d_{AC1,z} & d_{AC1,y} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\tan \beta}{k} & 1 & 0 & 0 \\ -d_{AC1,z} & 0 & d_{AC1,x} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\tan \alpha}{k} & 0 & -1 & 0 \\ -d_{AC1,y} & d_{AC1,x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{k} & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & d_{BC1,z} & d_{BC1,y} & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -d_{BC1,z} & 0 & -d_{BC1,x} & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -d_{BC1,y} & d_{BC1,x} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & v_{C2C1,x} & v_{C2C1,y} & v_{C2C1,z} \end{bmatrix} \quad (33)$$

The known terms vector $\mathbf{b}$ contain the external forces and the moments they generate, calculated with respect to the equivalent pole $\mathbf{C1}$:

$$\mathbf{b} = \begin{bmatrix} -X \\ -Y \\ -Z \\ -Z \cdot d_{DC1,y} - Y \cdot d_{DC1,z} \\ -X \cdot d_{DC1,z} - Z \cdot d_{DC1,x} \\ X \cdot d_{DC1,y} - Y \cdot d_{DC1,x} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (34)$$

The linear system was solved numerically in the MATLAB environment. To obtain the vector of unknowns, the left matrix division operator ($\backslash$) was used, which efficiently applies Gaussian elimination algorithms to solve the system $\mathbf{A} \cdot \mathbf{x} = \mathbf{b}$:

```
% Resolution of the linear system in MATLAB
Forces = Matrix \ b;
```

From the resulting vector **Forces**, the individual reaction vectors at the Force in B were extracted.

$$\mathbf{F}_b = \begin{bmatrix} 1.40e + 04 \\ -5.83e + 03 \\ 3.12e + 02 \end{bmatrix} \quad (35)$$

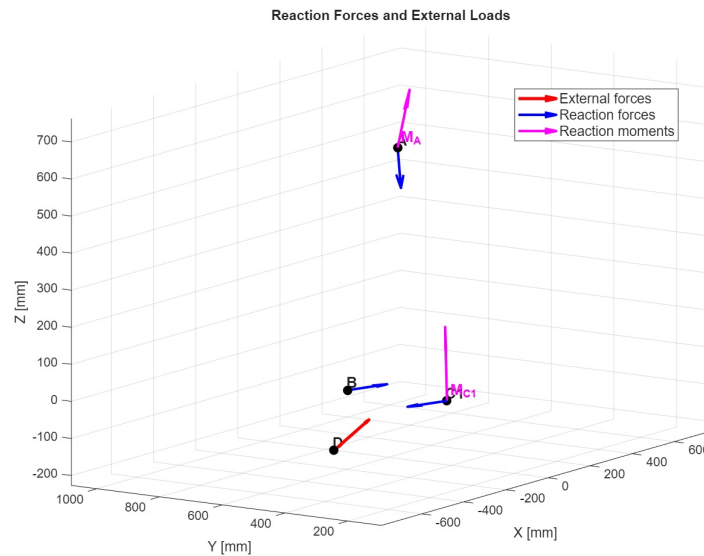


Figure 50: Free Body Diagram showing reaction forces, reaction moments, and external loads. Due to the significant difference in orders of magnitude among the vectors, their lengths are represented using a logarithmic scale for visual clarity.

6.4 Assembly components

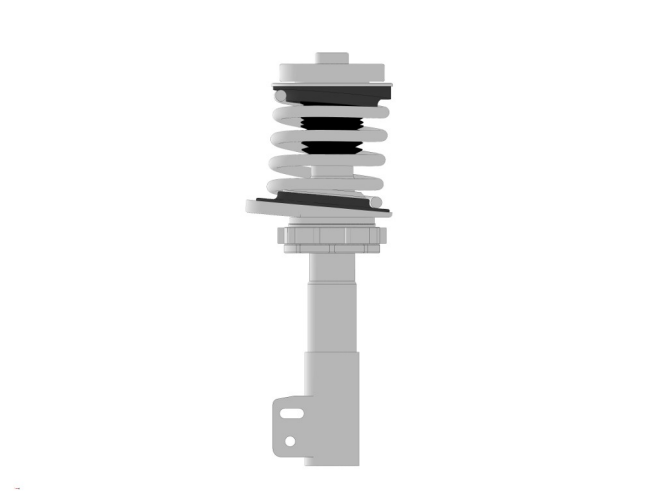
In this section, all the main components used to realize the suspension system are analyzed, including the parts embedded in the unsprung mass of the quarter car model.

The description goes deeper into the production phase, touching relevant topics such as the choice of the manufacturing process and the material, and other important features regarding functional and mechanical aspects of the parts.

6.5 Spring-damper assembly

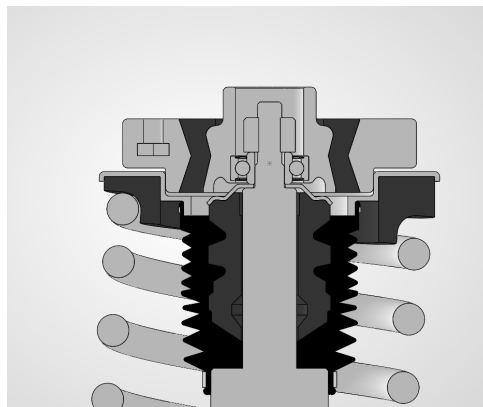
6.5.1 System Overview

The designed assembly is a variable-geometry spring-damper unit tailored for high-performance applications. The architecture is modular, allowing for fine-tuned kinematic adjustments, load decoupling, and reliable operation under dynamic stress.



6.5.2 Upper Interface & Top Mount Assembly

The top mount is engineered to provide both structural integrity and articulation:



- **Pseudo-Spherical Joint:** An SKF 6201-2Z bearing is embedded within a custom rubber isolator. This design effectively simulates the articulation of a spherical joint while dampening high-frequency NVH (Noise, Vibration, and Harshness).

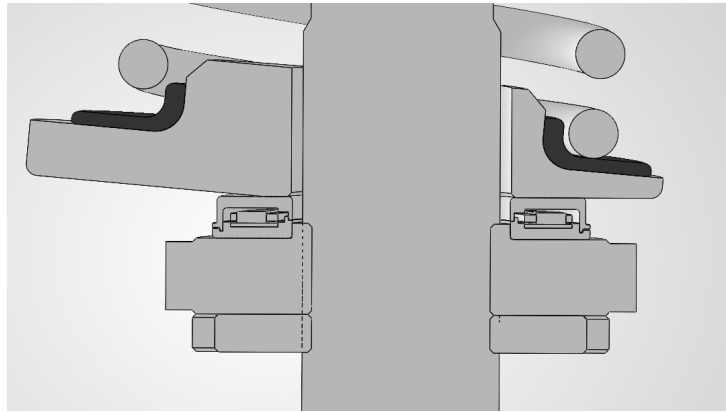
- **Rubber Retention:** To prevent the rubber isolator from dislodging under high vertical loads, the metallic housing features a convergent-divergent (hourglass) cross-section. Additionally, a high surface roughness was applied to the inner walls to maximize mechanical grip and rubber adhesion.
- **Fastening Strategy:** The main shaft and the upper bearing are secured using an all-metal prevailing torque nut (Stover-style, without nylon inserts) to guarantee thermal and vibrational reliability.
- **Angular Compensation:** To accommodate the geometric offset between the spring axis and the damper axis, a custom variable-thickness spacer (misalignment wedge) is integrated into the upper perch. This allows the spring to seat perfectly flush despite the 5° inclination.

6.5.3 Damper Body and Protection Systems

- **Travel Management:** An internal elastomer bump stop is mounted on a dedicated upper plate to safely manage bottom-out events and provide a progressive increase in the spring rate at the end of the stroke.
- **Environmental Sealing:** A flexible rubber dust boot is secured to the main damper body, fully enclosing the shaft and bump stop to prevent the ingress of abrasive dirt, dust, and moisture.

6.5.4 Lower Interface

The lower section of the assembly handles dynamic geometric changes and torsional loads:



- **Scrubbing Compensation:** As the suspension compresses, the geometric inclination generates lateral displacement (scrubbing). To accommodate this without binding, the lower spring perch features an engineered oval shape. Based on a standard 100 mm stroke, the lateral displacement Δx is calculated as:

$$\Delta x = 100 \cdot \tan(5^\circ) \approx 8.7 \text{ mm} \quad (36)$$

This custom geometry successfully manages the 8.7 mm of lateral travel without interference.

- **Spring Isolation:** A tailored rubber gasket covers the lower perch. This significantly increases friction between the spring coil and the perch seat—preventing unwanted shifting—while completely eliminating metal-to-metal contact and associated noises.
- **Torsional Decoupling (Thrust Bearing):** During compression, a coil spring naturally attempts to twist. If restricted, this adds parasitic friction, alters the effective spring rate, and causes binding. To solve this, the lower perch rests on an SKF AZK 6590 cylindrical roller thrust bearing, housed within a custom-designed retaining cage. This bearing completely decouples the spring’s rotation from the damper shaft, ensuring linear and bind-free compression.

6.5.5 Height Adjustment and Knuckle Mounting

- **Locking Mechanism:** Ride height is controlled via a threaded adjustment collar (spring seat) resting below the thrust bearing. To maximize reliability and prevent loosening under vibration, a secondary locking collar (jam nut) is utilized.
- **Hub Carrier Connection:** The lower damper body slides into a robust mounting sleeve (clevis). This sleeve features two mounting ears designed to bolt to the hub carrier. The upper mounting point uses a circular hole, while the lower point features a slotted hole to allow for precise camber adjustments.

6.6 Wheel rim

The wheel rim is an assembly of two distinct components: the **outer rim** and the **wheel center (disk)**.

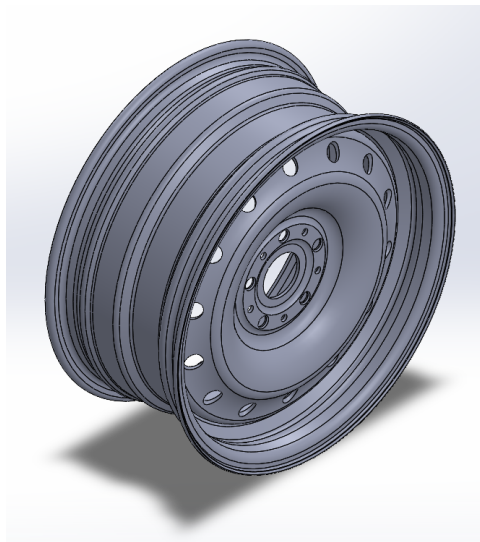


Figure 51: Wheel rim

The manufacturing sequence is defined as follows:

1. **Calendering and Welding.** A metal sheet is calendered to form an axisymmetric profile of constant thickness, followed by a weld to finalize the outer rim geometry.
2. **Deep Drawing** to form the inner disk.

3. **Punching** to create the holes for the bolts and the ventilation apertures.
4. **Press fitting and welding** to join the parts.

The chosen material is a **low-alloy steel**, selected for its exceptional balance of workability and weldability. This material provides high tensile and fatigue strength and good lightweight performance.

A particular feature of the rim is represented by the holes for the entrance of the bolts, which are characterized by a **tapered seat** that confers preload on the bolts, thus preventing unscrewing. Furthermore, this kind of seat behaves like a disk spring with low stiffness, which guaranties a better distribution of the forces on the screw enhancing the fatigue life.

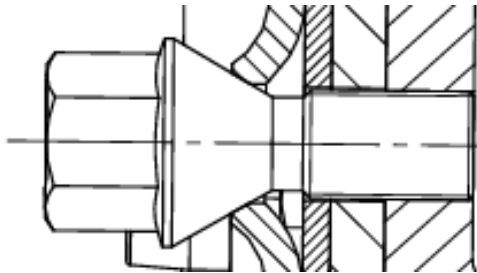


Figure 52: Conical screw and tapered seat

6.7 Brake disk

The brake disk was produced according to the following phase:

1. **Sand casting**, since the final shape is complex.
2. **Turning** to realize the central hole.
3. **Drilling** to create the holes for the bolts.

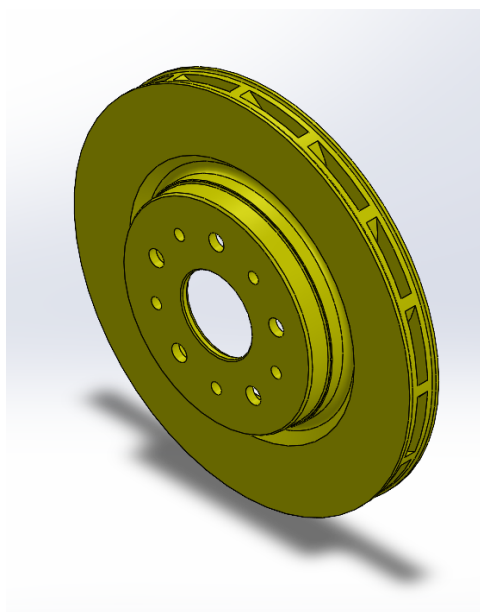


Figure 53: Brake disk

The chosen material is **cast iron**, picked for its thermal conductivity for efficient heat dissipation, good wear and compression resistance.

An important geometrical characteristic is shown below.

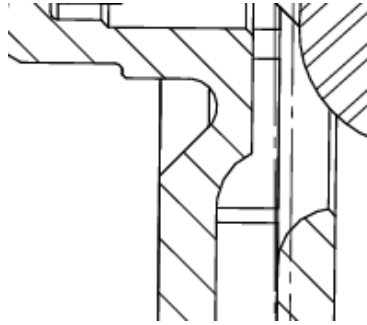
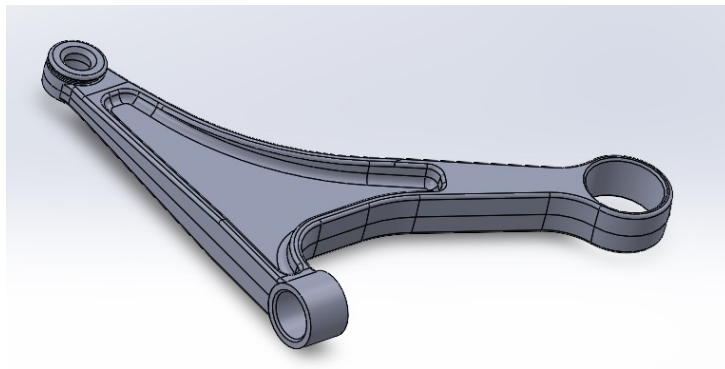


Figure 54: Fillet

This fillet is essential for the integrity of the disk when subjected to thermal expansion, otherwise the disk would bend on the wall of the hat.

6.8 Lower Arm

The design process of the Lower Control Arm follows a **Simulation-Driven Design** approach: starting from the kinematic requirements, passing through topological optimization to define the load paths, and concluding with a manufacturable CAD embodiment and FEM validation.



6.8.1 Geometric Constraints and Packaging

The preliminary design space is defined by the geometric constraints and the available packaging space within the vehicle's front corner.

- **Wheel Envelope and Steering Interference:** The most critical design constraint is the interference with the wheel rim and tire. The control arm operates within the confined volume of the wheel rim, and it must maintain safe clearance not only in the static position but throughout the entire range of motion.

The arm profile is shaped to accommodate a steering angle of 37° inward (toe-in) and 32° outward (toe-out) without contacting the rim or the brake disc.

6.8.2 Material and Manufacturing Process

To achieve the targets of mass reduction and high stiffness-to-weight ratio, Aluminum Alloy is selected as the structural material.

- **Production Process:** The component geometry is designed for High-Pressure Die Casting. This process allows for the creation of complex shapes with thin walls (I-beam web) and integrated housing for bushings, ensuring high repeatability and reduced machining costs.
- **Material Properties:** The design targets a casting alloy with a reference Yield Strength of ≈ 230 MPa and an Ultimate Tensile Strength of ≈ 300 MPa.

6.8.3 Bushing and Joint Configuration

The joint layout is designed to decouple handling precision from ride comfort:

- **Front Bushing (C1) – Horizontal Axis:** This bushing is designed to be radially stiff. Its primary role is to maintain the suspension geometry under high lateral loads (cornering) and braking forces, ensuring precise steering response and camber control. The high radial stiffness minimizes unwanted toe changes.

The connection to the subframe is a "double shear" type. The bushing is inserted into a U-bracket welded to the chassis subframe and secured with a horizontal through-bolt. The clamping force on the inner sleeve prevents any rotation of the sleeve relative to the chassis.

- **Rear Bushing (C2) – Vertical Axis:** This bushing acts as the primary compliance element for ride comfort. It features an anisotropic rubber structure.

The connection is a vertical bolted joint. The bushing rests against a flat reinforced surface on the chassis subframe and is clamped from below using a large washer and a vertical bolt.

6.8.4 Elastokinematic Analysis

A linear elastokinematic analysis is conducted to evaluate the compliance of the lower control arm under critical loading conditions.

The study utilizes a planar 3-DOF stiffness matrix model, decoupling the rigid-body equilibrium from the elastic deformation. The external loads at the ball joint (B), projected into the local arm coordinate system (u, v), are balanced against the reaction forces generated by the bushings at points C1 and C2.

To achieve the target separation of functions, anisotropic stiffness values are assigned:

Bushing	Position	K_u [N/mm]	K_v [N/mm]
Front Bushing (C1)	Horizontal Axis	2,000	15,000
Rear Bushing (C2)	Vertical Axis	1,500	10,000

The coordinate u represents the longitudinal direction (vehicle X-axis), while v represents the lateral/radial direction (vehicle Y-axis).

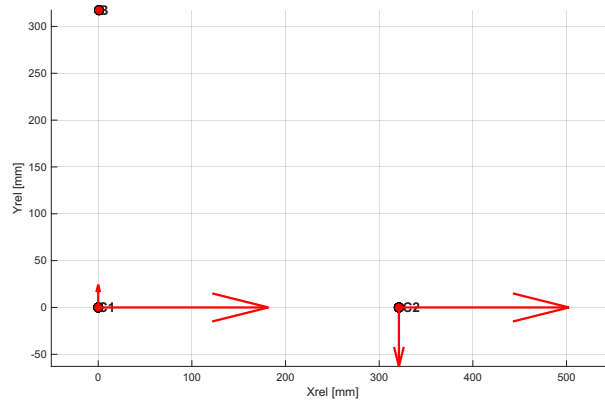
From a mechanical perspective, this configuration allows the bushings to be treated as **springs in parallel** regarding the longitudinal load. The total longitudinal stiffness of the system ($K_{tot,u}$) resisting the braking force is simply the algebraic sum of the individual stiffnesses:

$$K_{tot,u} = K_{u,C1} + K_{u,C2}$$

This parallel arrangement ensures that the total recession load is distributed between C1 and C2 proportionally to their respective stiffnesses, while maintaining a unified kinematic recession for the wheel center.

Under the calculated local loads, the system results in the following displacements:

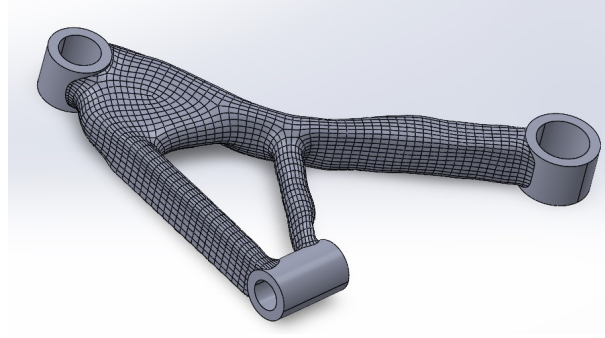
Parameter	Symbol	Value	Unit
Longitudinal Recession	u	4.00	mm
Lateral Deflection at Front Bushing	v_1	0.53	mm
Lateral Deflection at Rear Bushing	v_2	-1.39	mm
Arm Rotation (Yaw)	θ	-0.006	rad
Arm Rotation (Degrees)	θ_{deg}	-0.34	deg



6.8.5 Topological Optimization

Before defining the final CAD geometry, a topological optimization is performed using Altair Inspire to identify the optimal material distribution.

- **Objective:** Maximize Stiffness
- **Constraints:** Mass reduction target (30-50% of the initial design space)
- **Loads:** The optimization considers combined loading conditions (braking, cornering, and vertical bump) derived from the static analysis.



The software generates a truss-like structure, highlighting that the optimal load paths connect the wheel carrier (Point B) to the chassis mounts (C1 and C2) in a triangular formation, leaving a void in the center to save weight.

6.8.6 Embodiment Design

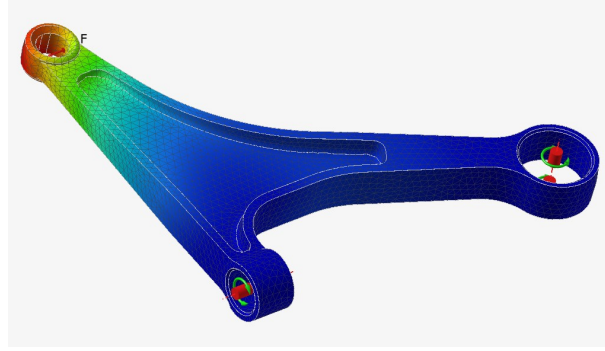
The organic shape resulting from the optimization is reconstructed into a manufacturable CAD model using SolidWorks. The design features are dictated by structural efficiency and casting requirements.

- **I-Beam Section:** The arm features a variable I-beam cross-section. This places the material at the extremities (top and bottom flanges) to maximize the moment of inertia (I) and bending stiffness, while a thin central web (4-6 mm) connects them.
- **Draft Angles:** To ensure manufacturability via the Split Draw process, a 3° to 4° symmetric draft angle is applied to all vertical walls relative to the middle parting plane.
- **Fillets and Radii:**
 - **Edges:** All external and upper edges are filleted with R3 mm.
 - **Internal Pockets:** Large fillets are used at the bottom of the pockets (wall-to-floor transition) to reduce stress concentrations and assist metal flow.

6.8.7 Structural Verification (FEM Analysis)

A linear static analysis is performed to validate the arm's structural integrity under combined loading conditions.

- **Simulation Setup:** The analysis utilizes standard isotropic aluminum properties (Young's Modulus $E \approx 73$ GPa, Poisson's ratio $\nu = 0.33$).
- **Displacement Results:** The maximum deflection at the ball joint is 1.12 mm corresponding to point B.

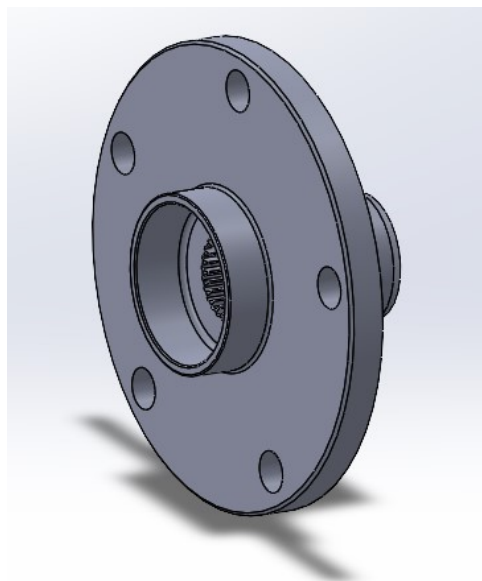


- **Safety Factor Verification:** Comparing the calculated stress ($\sigma_{max} = 115$ MPa) against the reference yield strength of a standard casting alloy ($\sigma_{yield} \approx 230$ MPa), the component achieves a Safety Factor of ≈ 2.0 in the most stressed region.



6.9 Hub

The wheel hub serves as the primary rotating structural interface within the MacPherson suspension assembly. Its main functions are to support the vehicle's weight, manage dynamic loads (vertical, radial, and axial forces during cornering and braking), and smoothly transmit drive torque from the drivetrain to the wheels while maintaining precise alignment.



The hub assembly integrates with the following key elements:

- **Wheel Bearing:** A double-row ball bearing is mounted on the hub's outer diameter to ensure low-friction rotation while handling combined radial and axial loads.
- **CV Joint (Drive Shaft):** Interfaces with the hub via an internal splined connection and is rigidly secured by a central axle nut to deliver torque.
- **Brake Disc:** Sandwiched against the hub's outer flange, providing a stable surface for the braking system.
- **Wheel Fasteners:** This hub utilizes a wheel bolt fastening system, a configuration commonly adopted by European automotive manufacturers.

Instead of using a stud and lug nut system where threaded studs permanently protrude outwards from the hub, this hub's flange features threaded through-holes. The wheel bolts are inserted from the front, passing through the wheel rim and the brake rotor hat, to thread directly into the hub.

To prevent misalignment issues between the parts to be assembled, the design requires a rotor set screw or an assembly guide pin. This pin locks the brake disc to the hub when the wheel is dismantled and provides a convenient resting point for the user to hang the wheel.

6.10 Hub Bearings

For the front McPherson suspension, the design objective is to replicate the architecture of a **Generation 1 Wheel Hub Unit (HUB1)**. This configuration consists of a double-row angular contact ball bearing and represents the automotive industry standard for compact vehicles due to its ability to handle combined loads (radial, axial, and tilting moments) within a compact envelope.

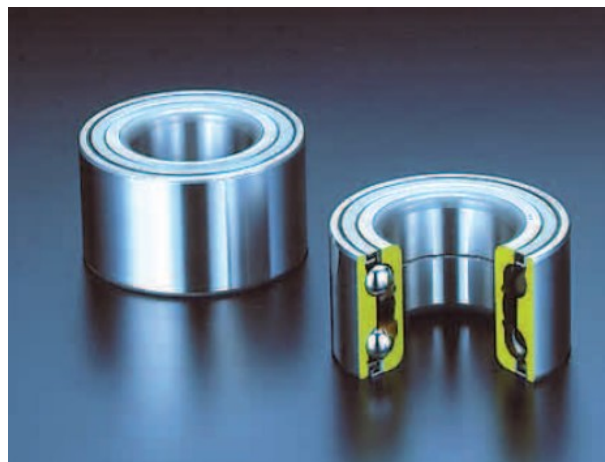


Figure 55: HUB1 bearing

The defining feature of the HUB1 is the split inner ring:

- **Geometric Preload:** Unlike standard industrial bearings, the HUB1 relies on "geometric preload." When the unit is loose, there is internal clearance. The de-

sign preload (negative clearance) is achieved only when the two inner half-rings are clamped together until they touch.

- **Assembly Requirement:** This clamping is provided by the tightening torque of the CV joint nut.
- **Mounting Configuration:** A critical design choice is the internal arrangement of the contact angles. The selected bearing utilizes a **”O” configuration**.

In this arrangement, the lines of action of the pressure forces diverge towards the bearing axis, creating a large distance between the pressure centers ($2a$), effectively wider than the physical bearing width.

6.10.1 ”Hybrid” Selection Strategy:

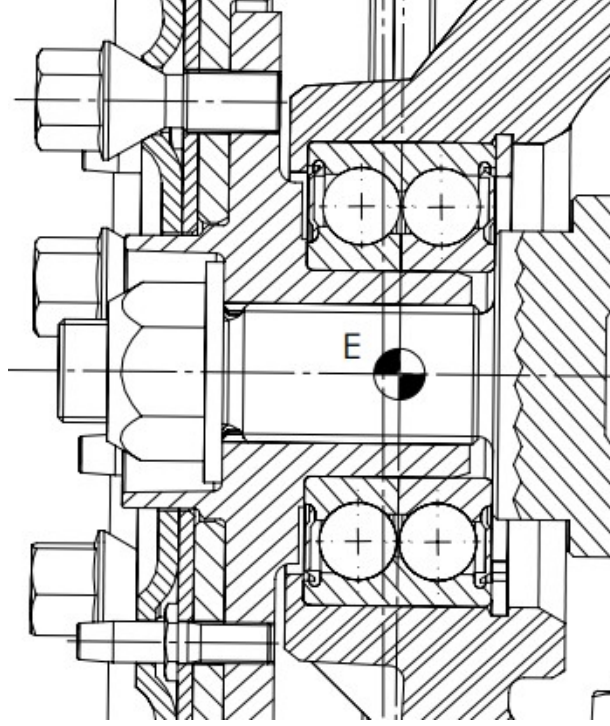
While automotive-specific catalogs exist, they often lack the detailed calculation coefficients required for academic verification. Therefore, an industrial equivalent from the **SKF General Catalog** is selected to ensure access to validated data (C, C_0, e, Y) for structural sizing.

To resolve this, a ”Hybrid” definition is adopted, distinguishing between the physical calculation model and the CAD representation:

- **Calculation Model: SKF 3308 DMA** This bearing is selected because it features a Split Inner Ring. This is the only architecture that respects the true kinematics of a wheel hub, where preload is not factory-set but generated during assembly.

Parameter	Symbol	Value	Unit/Note
Designation	-	3308 DMA	Series 33 D
Arrangement	-	Back-to-Back	”O” Configuration
Contact Angle	α	≈ 45	deg ($^{\circ}$)
Inner Diameter	d	40	mm
Outer Diameter	D	90	mm
Width	B	36.5	mm
Load Center Distance	a	84	mm
Dynamic Load Factor	C	68.9	kN
Static Load Factor	C_0	57.0	kN
Fatigue Load Limit	P_u	2.45	kN
Limiting Speed	n_{lim}	7000	r/min
Calculation Factor	e	1.34	Threshold
Dynamic Factor 1	Y_1	0.47	For $F_a/F_r \leq e$
Dynamic Factor 2	Y_2	0.81	For $F_a/F_r > e$
Static Factor	Y_0	0.44	-

- **CAD Model: SKF 3308 A-2RS1** This model is used for the geometric design because it features integrated seals, which are essential for protecting the hub from environmental contaminants. The 3D model is then modified to visually divide the inner ring, making it functionally identical to the HUB1 series.



6.10.2 Structural Verification and Sizing:

The loads are calculated based on the "Worst Case" scenario for static verification, and a nominal load case for fatigue.

A critical step in the analysis of an HUB1 is resolving the Overturning Moment (M) generated by the offset between the wheel center and the tire contact patch.

The global forces and moment acting on the hub center (E) are:

$$F_{rad.total} = \sqrt{X^2 + Z^2}$$

$$F_{axial} = |Y|$$

$$M = |(Y \cdot \Delta z) - (Z \cdot \Delta y)|$$

To determine the load on the most stressed row of the bearing, the moment M is converted into an additional radial load. The equivalent radial load (F_r) on a single row is calculated as:

$$F_r = \frac{F_{rad.total}}{2} + \frac{M}{2a}$$

- **Static Verification:** The static safety factor is calculated to ensure no permanent deformation occurs under peak loads.

$$P_0 = X_0 \cdot F_r + Y_0 \cdot F_{axial}$$

The verification compares the load on a single row against the static capacity of a single row.

$$\eta_{static} = \frac{C_0/2}{P_0}$$

With $C_0 = 57$ kN, the calculated $\eta_{static} > 1$, validating the choice of the 40mm bearing (SKF 3308 DMA).

- **Fatigue Verification:** Fatigue life (L_{10}) is estimated using nominal loads

$$P = X \cdot F_r + Y \cdot F_{axial}$$

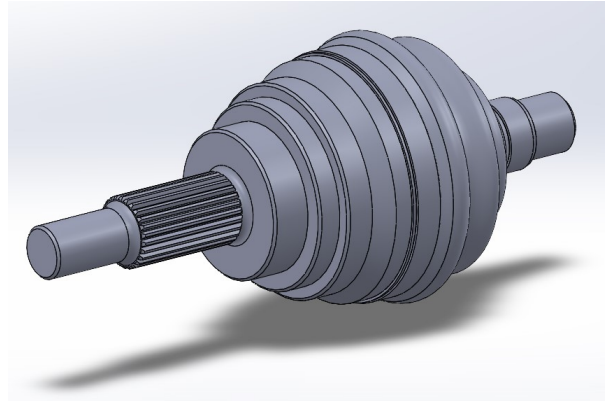
$$L_{10} = \left(\frac{C/1.62}{P} \right)^3 \quad [\text{millions of revolutions}]$$

The calculated life confirms the bearing's durability for the required vehicle mileage under mixed driving conditions.

6.11 CV joint

6.11.1 Working Principle

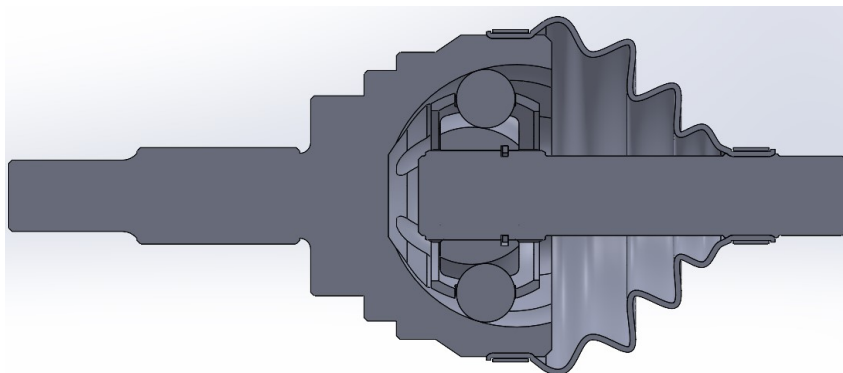
The primary function of the CV joint is to transfer torque from the drivetrain to the wheels at variable angles while maintaining a constant rotational speed. Unlike a standard Cardanic joint, which induces speed fluctuations and vibrations when operating at significant angles, the CV joint ensures smooth power delivery and reduced wear, even during sharp steering maneuvers and extensive suspension travel.



6.11.2 Design Choices and Mounting

The design features a fixed-ball configuration (Rzeppa style) characterized by a spherical outer race and an inner race enclosed within a cage.

To secure the drive shaft to the inner race, a ring sits within a rectangular groove. This locks the shaft axially, preventing separation at the wheel hub. Notably, while this outboard joint remains axially fixed, the corresponding inboard joint (chassis side) allows for axial movement to accommodate shaft length changes caused by suspension geometry.

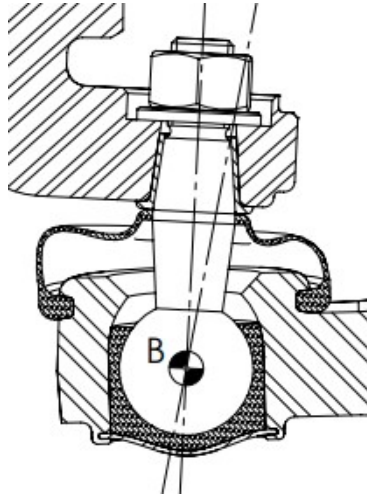


The connection between the CV joint and the wheel hub utilizes a 26-tooth involute spline profile.

6.12 Spherical Joint

6.12.1 Working principle:

The ball joint serves as the critical pivot point connecting the suspension Lower Control Arm to the Hub Carrier. It manages the articulation required for suspension travel and steering while transferring lateral and longitudinal loads.



6.12.2 Mounting features:

- **Lower Arm Assembly (Housing Interface):** The ball joint housing acts as an integral component of the lower arm. The internal mechanism is assembled from the underside.

The sphere is immersed in a low friction material (polymeric) to allow the rotation of the sphere in the housing.

The bottom closing plate is secured via a crimping process. The lower lip of the arm's housing is plastically deformed over the plate to seal the internal components and ensure structural integrity.

This mechanism requires some adjustments in the production process of the lower arm itself: the lower arm is cast with some machining allowance (extra stock), as the thin retention lip cannot be cast directly due to tolerance limits. The vertical lip is subsequently **CNC machined** from solid to guarantee perfect concentricity with the housing.

- **Hub Carrier Interface (Stud Connection):** The connection to the hub carrier utilizes a tapered stud secured by a top nut.

A conical sleeve (bushing) is interposed between the ball stud and the hub carrier bore. This component is critical for material compatibility (typically mating a steel stud to an aluminum carrier) and wear reduction.

- **Environmental Protection:** The assembly is protected by a flexible rubber dust boot secured at both the stud and housing neck. This seal prevents the ingress of contaminants (water, grit) and retains the lubricating grease, ensuring the longevity of the joint.

6.13 Hub Carrier

The hub carrier is a part that connects the hub with the damper, the tie rod and the lower arm.

It is produced via **low pressure die casting (LPDC)** due to the complex shape. This process guarantees high mechanical performances and a nice structural integrity, since it prevents the formation of porosity and voids.

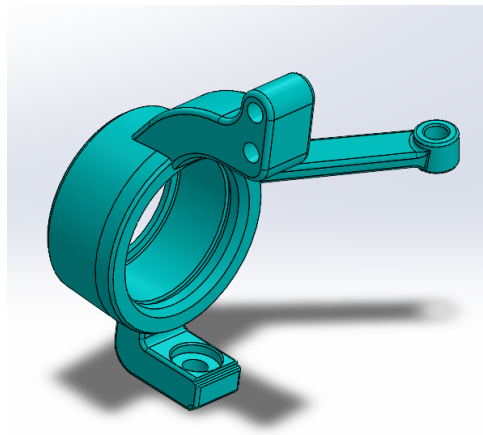


Figure 56: Hub carrier

The chosen material is a **cast aluminum alloy**, used to reduce the weight of the unsprung mass, thus enhancing suspension performance in terms of comfort and road holding.

7 Visit to DriSmi

7.1 Introduction

This chapter focuses on the visit at **DiM400 DYNAMIC Driving Simulator by VI-grade** of Politecnico di Milano. This facility bridges the gap between virtual modeling and real-world testing. The visit highlighted how this technology enables researchers to evaluate vehicle performance, safety systems, and human-machine interaction in a controlled, highly immersive environment.

7.2 Characteristics of the simulator



Figure 57: Driving simulator

The simulator is employed to assess and perform studies on:

- integration of electronic control units;
- resistance and durability on mechanical components;
- ride and handling characteristics;
- real driving emissions (RDE);
- Advanced Driver Assistance Systems (ADAS) and fully autonomous vehicle protocols.

The DiM400 is a mid-size dynamic driving simulator with the following key features:

- a 6 m x 6 m platform, which provides a large workspace for dynamic movements;
- cables used to move the diskframe, offering more workspace compared to conventional electric actuators.

The diskframe's movements are independent across the platform, offering 3 degrees of freedom (DOFs) and a bandwidth of approximately 3 Hz. The support provided by 3 airpads enables frictionless sliding of the diskframe on the platform.

Furthermore, the presence of the Hexalift adds 6 additional DOFs to the cockpit, with a bandwidth of approximately 30 Hz. Other devices include:

- 8 onboard shakers reproduce engine vibrations and road irregularities, achieving a bandwidth of around 200 Hz;
- an Inertial Compensation System (ICS) ensures stability and precision in the simulator's movements;
- active belts/active seat mimic the effects of sustained longitudinal/lateral accelerations;
- active steering for a realistic feedback and simulate active steering control systems;
- active brake to reproduce the proper pedal feeling and the effect of active controls like ABS;
- 5 speakers inside the cockpit reproduce the noise sources inside and outside the vehicle.

7.3 Tested maneuver

The test involved driving at a constant speed over a series of equally spaced bumps. The goal was to analyze the vertical dynamic response of the vehicle and distinguish the oscillations of the unsprung masses, which include the wheels, tires, and suspension links, from those of the sprung masses, primarily the vehicle body. The maneuver was run first at velocity capable of exciting the sprung mass natural frequency (1-2 Hz), then the same but for the unsprung masses (10-13 Hz).

According to the quarter-car vehicle model, it is possible to define the resonating frequency of both the sprung and unsprung masses, of which the transfer function between the road displacement and the mass displacement is thereafter depicted:

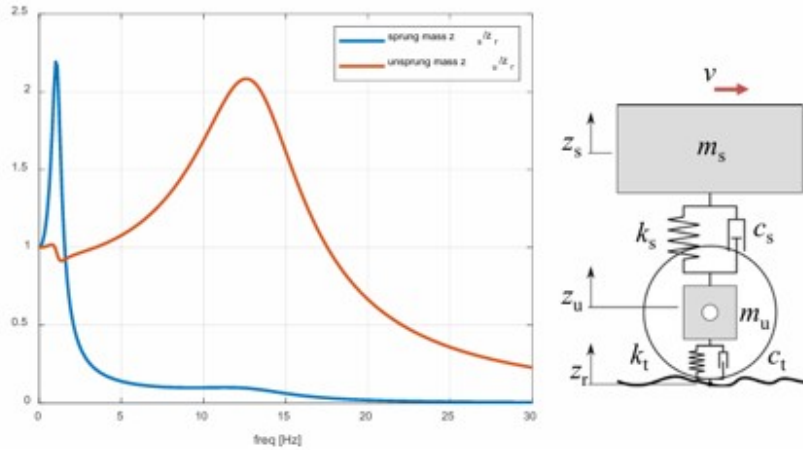


Figure 58: Sprung and unsprung masses transfer functions

It is possible to mathematically define discomfort as the ratio between the root means square of the sprung mass acceleration $\ddot{z}_s$ and that of the road irregularity $\ddot{z}_r$. The lower this value, the more comfortable the vehicle:

$$Discomfort = \frac{RMS[\ddot{z}_s]}{RMS[\ddot{z}_r]} \quad (37)$$

Then the transfer function of the discomfort was computed to evaluate at which frequency the discomfort reaches its maximum value.

As shown in the following picture, the greatest discomfort is felt when the unsprung masses were excited into resonance.

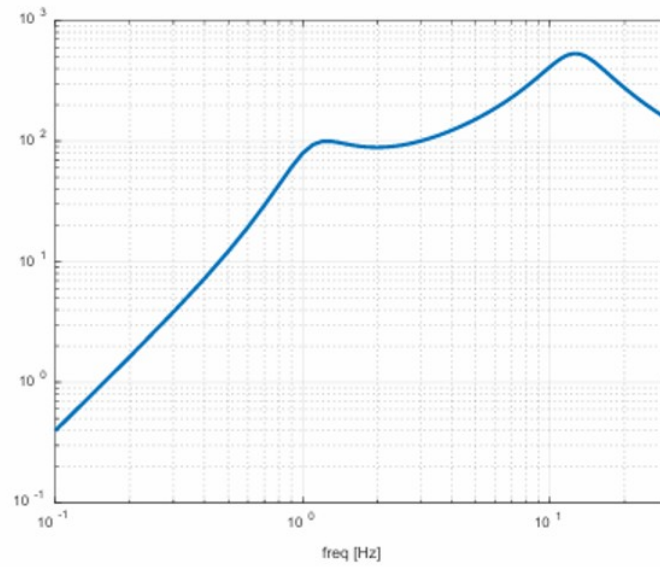


Figure 59: Discomfort TF

8 Visit to LaST

8.1 Introduction

This section is dedicated to the visit at **Laboratorio sulla Sicurezza dei Trasporti (LaST)** of Politecnico di Milano. The aim was to analyze the key mechanical systems of cars, focusing the attention on their design and the functionality of their main components.

8.2 Mechanical systems studied inside the lab

Inside the structure, the students had the opportunity to have a look to two different types of suspensions: the MacPherson and the multi-link suspension.

The **MacPherson suspension system** is a compact setup composed by a single lower control arm, a coil spring, and a damper integrated into a single unit. It is largely used in passenger vehicles for its economic cost compared to the other systems and its small size.



(a) Front suspension



(b) Rear suspension

Figure 60: MacPherson suspension systems

The **multi-link suspension** is a more advanced system, consisting of multiple individual links connecting the wheel hub to the vehicle chassis. This configuration allows for precise control of wheel alignment parameters, such as camber and toe, throughout the suspension travel. It is widely used in high-performance and luxury vehicles for its versatility in optimizing ride comfort and handling.

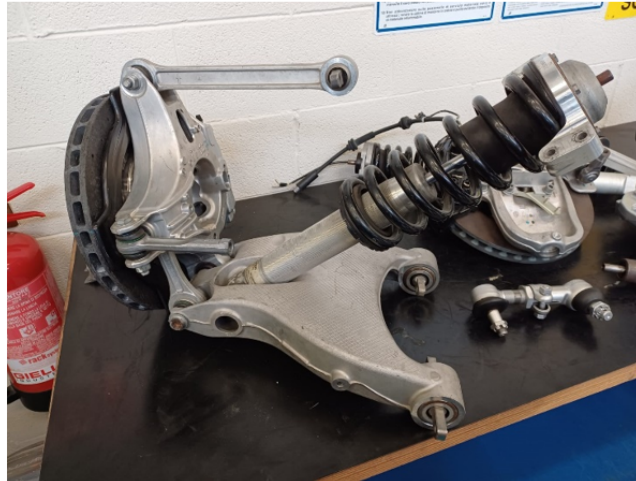
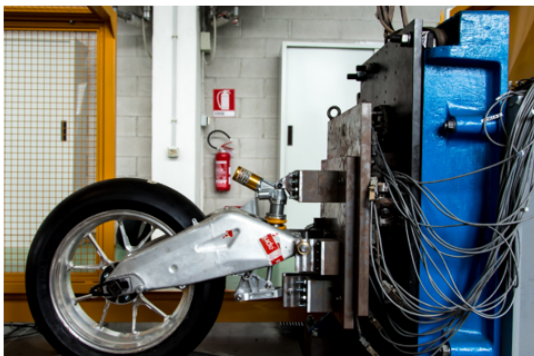


Figure 61: Multi-link suspension

Other important devices were observed, including:

- a **carbon-ceramic brake disc** with drilled holes. The material properties of the disc, such as high thermal resistance, low weight, and durability, make it a preferred choice in high-performance and racing applications. The drilled holes were noted as features to enhance heat dissipation;
- an **active differential**, which is able to actively vary torque distribution between wheels to enhance traction and handling. Unlike a traditional open differential, the active variant uses electronically controlled clutches or other mechanisms to dynamically adjust torque, adapting to changing road and driving conditions;
- a **clutch assembly**, including its pressure plate, friction disc, and release mechanism. The clutch's role in transmitting torque from the engine to the transmission while allowing for smooth engagement and disengagement.

Finally, tire testing setups such as the **drum** and the **Moving Laboratory for Automotive components Safety testing – MoLAS** were showed, and their functions, for example the identification of Pacejka formula's parameters, were discussed. A testing machine for bicycle tyres was also presented.



(a) Drum test



(b) MoLAS

Figure 62: Tyre testing setups

9 Cromodora seminar on aluminum wheels

9.1 Introduction

This chapter is a review of a seminar about aluminum alloy wheels held by Cromodora, focusing on the main features about manufacturing phases, testing procedures and production management.

9.2 Manufacturing processes

According to the manufacturing process, aluminum wheels are divided in two categories: **cast** and **forged** wheels.

The choice of the process depends on the final required objectives to be reached in terms of cost, performance and styling.

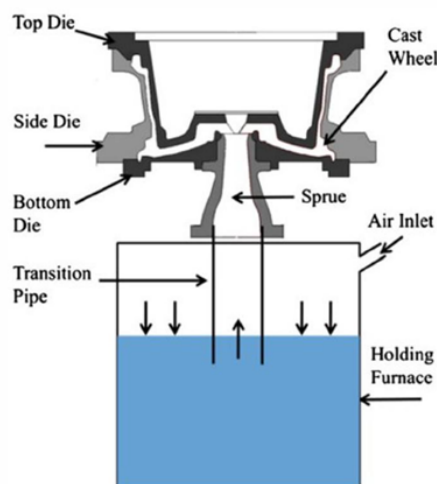
9.2.1 Cast wheels

Cast wheels can be either obtained from **low pressure (LPDC)** or **gravity die casting process (GDC)**. These are by far the most common manufacturing technologies in the market as they allow to produce an infinite variety of stylings with good material characteristics at a reasonable cost.

In **LPDC** the molten metal fills the die from the bottom through a transmission tube which dips into a lower holding furnace pressurized with air to push metal in the die.

The solidification process proceeds oppositely to the metal injection. To prevent porosity formation, the solidification speed must be low.

In **GDC** the molten metal is poured in the die, which is filled by gravity. The casting machine can tilt to improve the die filling.



(a) LPDC scheme



(b) GDC scheme

Figure 63: Casting processes

In both cases, once the die is full of metal the solidification is driven by the design of the cast part combined with heaters and cooling systems.

9.2.2 Forged wheels

Forged wheels have **better mechanical properties** and **enhanced lightweight**. On the other hand, the cost is higher compared with cast wheels and stylings are more complex to be realized.

These type of wheels are divided in two categories as well: **net forged wheels** and **blank forged and machined wheels**.

Net forging is meant for low cost and high volumes productions, but the variety of stylings is very limited.

The full process is composed by the following phases

1. Preliminary forging of a cast bar
2. Design forging
3. Perforation to obtain the styling
4. Flow-forming to elongate the wheel
5. Final finishing

Blank forging and machining is performed to obtain complex shapes, but this leads to a more expensive product. This process is realized for small batches. The wheel is manufactured from a metal billet forged and machined via milling.

9.3 Main features

During the design of a wheel, many aspects must be taken into account in terms of:

- **Geometrical features** defined by the car and wheel manufacturers. Elements like the rim width, the wheel diameter, the offset, the hump and rim type fall into this category.
In particular, the number of holes is a challenging choice. From one side a high number of holes improves casting process but it is preliminary determined by the car manufacturer for styling purposes.
- **Interface constraints.** The wheel must leave space for housing the tyre, the brake caliper, the hub and other parts. It must be compatible with bolt holes.
- **Aesthetic features.** Wheels are properly painted or milled for cosmetic purposes.
- **Dynamic features.** For testing purposes, the lateral and vertical loads are considered. An important activity is the computation of the natural frequencies of the wheel for Noise, Vibration, and Harshness (NVH) analysis
- **Functional features.** Wheels must comply with mechanical, metallurgical and corrosion test requirements. Mechanical tests are mainly checking the wheel resistance to rotary fatigue during cornering and to impacts.
Paint tests are checking the resistance of the painted layers into acid environments and in scratched conditions.
Metallurgical tests are checking the microstructure, the material characteristics (Rm, Rp, A%), the porosity and the chemical composition.

Functional tolerances are focused on defining a limit of the dynamic unbalance during the wheel rotation to which the same error in the tyre is added. The cumulated error must be then compensated with balance weights commonly used by all tyre fitters.

9.4 APQP

When a new wheel is launched, **Advanced Product Quality Planning** is used to organize the design and production activities according to five main phases, reported in the table below.

APQP Phase	Objective	Main activities
Phase 1	Project feasibility	Feasibility/cost analysis, timing plan
Phase 2	Product design	CAD modeling, FEA for structural validation, DFMEA, casting simulation
Phase 3	Production tooling and validation	Product tooling, sampling, mechanical tests
Phase 4	Product approval for production	Quality studies, analysis of production performance
Phase 5	Production ramp-up	MOP, production capacity installation

Table 12: APQP Phases and Responsibilities

Within the whole process, compulsory tests such as mechanical impact and fatigue test, paint resistance test to stone impact and metallurgical tests must be carried out.

Production must start within about one year from the design presentation.

10 Capgemini seminar

Capgemini is a global leader in consulting and technology services, playing a pivotal role in the automotive sector's transition towards Software Defined Vehicles (SDV). As vehicles become increasingly complex—integrating AI, IoT, and autonomous driving features—the boundary between mechanical engineering and software validation blurs.

The seminars highlighted a dual challenge for modern OEMs:

1. **Safety:** Ensuring that electronic systems do not fail in a way that harms people (Functional Safety).
2. **Reliability:** Guaranteeing that mechanical designs are robust and durable from day one to minimize "non-quality costs" (Preventive Reliability).

10.1 Part I: Functional Safety (ISO 26262)

Functional Safety has become a paramount concern in the automotive industry, driven by the increasing complexity of Electrical and Electronic (E/E) systems. A historical turning point was the **Toyota Accelerator Incident (2009)**, where unintended acceleration led to fatal accidents, highlighting the critical need for rigorous validation of electronic throttle controls.

Safety is formally defined as the "freedom from unacceptable risk of physical injury". It is crucial to distinguish between:

- **Passive Safety:** Systems that mitigate damage *during* an accident (e.g., Airbags, Seatbelts).
- **Active Safety:** Systems that prevent accidents from happening (e.g., ABS, AEB).
- **Functional Safety:** A subset of active safety that relies on the correct functioning of E/E systems to manage risks.

Since "Zero Risk" is impossible to achieve in real-world operations, the goal is to reduce risk to a **Tolerable Level** (ALARP principle). Key definitions include:

- **Harm:** Physical injury or damage to health.
- **Hazard:** A potential source of harm caused by malfunctioning behavior.
- **Residual Risk:** The risk remaining after safety measures are applied.

IEC 61508 is the "mother standard" for functional safety across all industries. However, the specific needs of the automotive sector led to the creation of **ISO 26262**.

ISO 26262 is structured into **12 Parts** (10 normative, 2 guidelines), covering the entire lifecycle:

1. Vocabulary.
2. Management of Functional Safety.
3. **Concept Phase:** Item definition and Hazard Analysis.
4. **Product Development (System Level):** System architecture and integration.

5. **Product Development (Hardware Level):** HW design and metrics.
6. **Product Development (Software Level):** SW safety requirements and testing.
7. Production, Operation, and Service.
8. Supporting Processes.
9. ASIL-oriented and Safety-oriented Analyses.
10. Guidelines (Informative).
11. Guidelines for Semiconductors.
12. Adaptation for Motorcycles.

The standard follows the V-Model workflow. The left side (descending) represents the design and refinement of requirements (System → HW/SW), while the right side (ascending) represents the integration and verification (Testing → Validation).

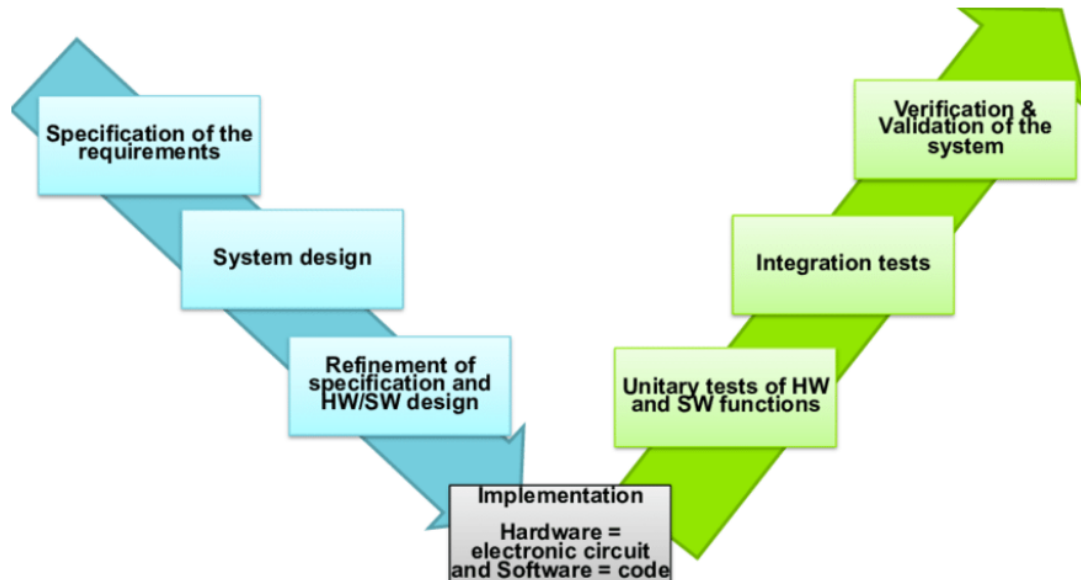


Figure 64 The ISO 26262 V-Cycle illustrating the relationship between development and testing phases.

The core of the Concept Phase is the **Hazard Analysis and Risk Assessment (HARA)**. This method identifies potential hazards and classifies them using the **Automotive Safety Integrity Level (ASIL)**.

The ASIL (ranging from QM to D) is determined by combining three parameters:

1. **Severity (S):** The extent of harm to individuals.
 - **S0:** No injuries.
 - **S1:** Light/moderate injuries.
 - **S2:** Severe/life-threatening injuries (survival probable).
 - **S3:** Life-threatening/fatal injuries (survival uncertain).

2. **Exposure (E):** The probability of being in the operational situation where the hazard can occur.
 - **E1:** Very low probability (e.g., jump start).
 - **E2:** Low probability ($< 1\%$ operating time).
 - **E3:** Medium probability ($1\% - 10\%$).
 - **E4:** High probability ($> 10\%$, e.g., heavy traffic, highway driving).
3. **Controllability (C):** The ability of the driver to avoid the accident.
 - **C0:** Controllable in general.
 - **C1:** Simply controllable ($> 99\%$ of drivers).
 - **C2:** Normally controllable ($> 90\%$ of drivers).
 - **C3:** Difficult to control or uncontrollable ($< 90\%$ of drivers).

Once the ASIL is assigned, the standard provides tables with recommended methods for validation. These entries can be:

- **Consecutive entries:** All methods listed must be applied (marked by 1, 2, 3...).
- **Alternative entries:** Only one or a combination of methods is required (marked by 1a, 1b...).

Methods		ASIL			
		A	B	C	D
1a	Environmental testing with basic functional verification ^a	++	++	++	++
1b	Expanded functional test ^b	o	+	+	++
1c	Statistical test ^c	o	o	+	++
1d	Worst case test ^d	o	o	o	+
1e	Over limit test ^e	+	+	+	+
1f	Mechanical test ^f	++	++	++	++
1g	Accelerated life test ^g	+	+	++	++
1h	Mechanical Endurance test ^h	++	++	++	++
1i	EMC and ESD test ⁱ	++	++	++	++
1j	Chemical test ^j	++	++	++	++

Figure 65 ASIL Determination Matrix combining S, E, and C parameters.

10.1.1 Case Study: Active Suspension System

The seminar applied HARA to an **Active Suspension System** powered by a high-voltage battery. The analysis identified the following Safety Goals (SG):

- **BCM_SG1 (Thermal Runaway):** Risk of fire on the Multi-Voltage Board.
 - *Assessment:* High Severity (S3), High Exposure.
 - *Result:* **ASIL C**.
 - *Safe State:* Cut off power flow.
- **BCM_SG2 (Electrocution):** Risk of human contact with high voltage.
 - *Result:* **ASIL B**.
 - *Safe State:* Discharge 48V bus to $< 60\text{V}$.
- **BCM_SG3 (Cable Fire):** Risk of fire on 48V cables.

- *Result:* **ASIL A.**
- *Safe State:* Switch off devices.

10.2 Part II: Preventive Reliability (D-FMEA)

Before detailing the specific methodology, it is crucial to understand the broader framework of Preventive Reliability. This approach is not a single step but a continuous integration of two main activities throughout the product lifecycle:

1. **FMEA (Failure Mode and Effect Analysis):** Applied during the development phase. It is proactive, aiming to predict and prevent failures before they occur.
2. **Problem Solving:** Applied during the experimental, production, and working phases. It is reactive, triggered when a failure actually occurs during testing or operation.

The combination of these two activities creates a closed-loop system. When a failure is detected and resolved through Problem Solving, the solution is not just a "quick fix" but is recorded. This generates a Database of Lessons Learned. This database is critical for future projects, ensuring that the same errors are not repeated in new designs, effectively transforming reactive experience into proactive wisdom.

The reliability process is deeply integrated into the standard Product Development Process, which follows a specific flow:

- **Start:** Concept Phase.
- **Development:** Design Development and Process Development run in parallel.
- **End:** Production and Start of Production (SOP).

Preventive reliability activities start from the very beginning (Concept) and continue through various checkpoints. A key aspect is the continuous monitoring of failure effects: feedback loops based on Problem Solving outcomes are constantly inserted back into the development process to refine the design.

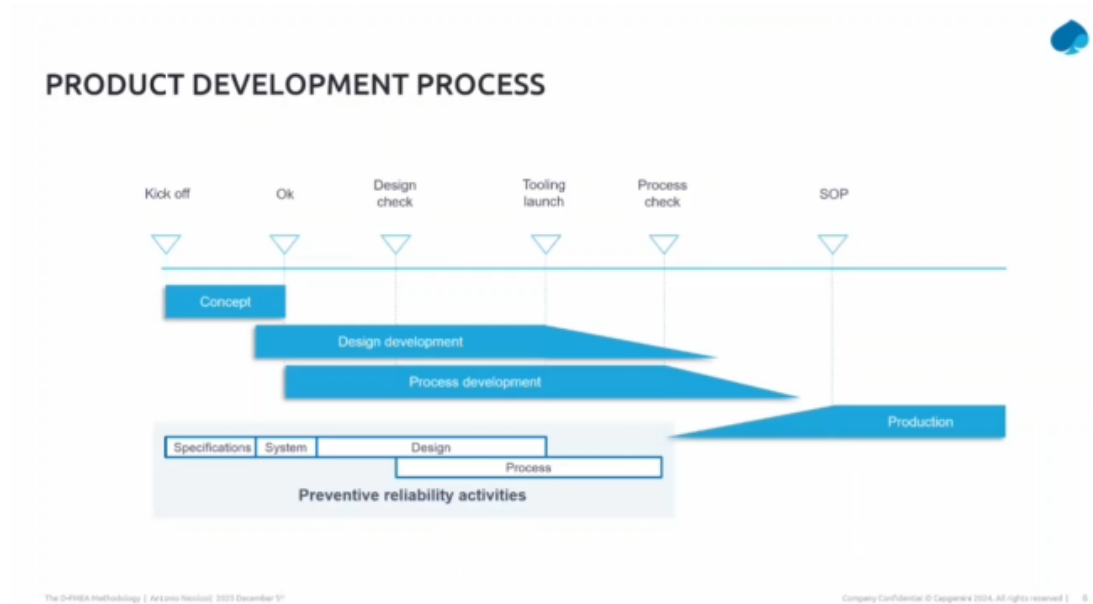


Figure 66 The Product Development Process: parallel development of design and process with continuous reliability integration.

One of the primary business drivers for Preventive Reliability is the reduction of Non-Quality Costs (NQC).

Definition: Non-Quality Costs represent the total financial loss resulting from product imperfections. They include Internal Failure Costs (scrap, rework before delivery) and External Failure Costs (warranty claims, recalls, legal liabilities).

A fundamental rule of reliability engineering is that NQC grows exponentially the later a failure is discovered. A defect found during the design phase costs little to fix (changing a drawing), whereas the same defect found during production or by the customer entails massive costs. The seminar presented a comparative analysis (histogram) of NQC across different approaches:

1. **Scenario 1: No Preventive Analysis.** This scenario results in the highest NQC. Without early detection, most failures are found late, leading to expensive "fire-fighting" and modifications.
2. **Scenario 2: Focus on New Content Only.** Here, preventive analysis is applied only to new components, while "carry-over" parts (parts reused from previous models) are ignored. The costs for carry-over parts remain high, resulting in moderate NQC.
3. **Scenario 3: Global Preventive Analysis.** This is the optimal scenario. By applying preventive reliability to the entire system (both new and carry-over parts), the organization achieves the lowest NQC, maximizing the "Lessons Learned" from previous projects.

THE RELIABILITY PLAN

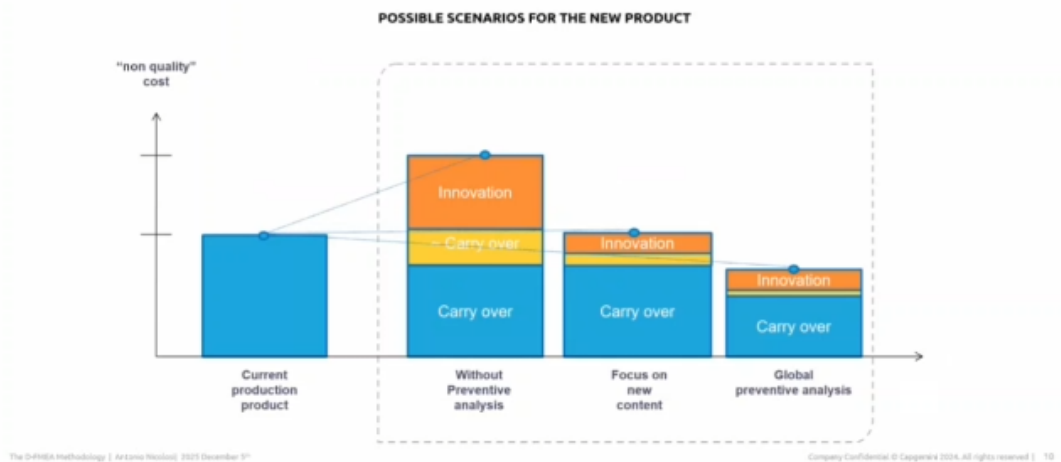


Figure 67 Comparison of Non-Quality Costs across different preventive strategies.

To achieve the "Global Preventive Analysis" mentioned above, the Design Failure Mode & Effect Analysis (D-FMEA) is the standard tool. It follows a structured 7-step process: System analysis:

1. **Planning & Preparation:** Defining the cross-functional team (Engineering, Testing, Quality, Manufacturing) and the scope.
2. **Structure Analysis:** Using Block Diagrams to map physical boundaries, elements, and interfaces.
3. **Function Analysis:** Using Correlation Matrices to map functions to specific elements.

Failure Analysis & Risk Mitigation:

4. **Failure Analysis:** Identifying the chain of failure (Cause → Mode → Effect) using the "5 Whys" method to find the root cause.
5. **Risk Analysis:** Calculating the Risk Priority Number (RPN) based on Severity, Occurrence, and Detection.
6. **Optimization:** Developing a corrective action plan to reduce the RPN.

Risk Communication:

7. **Results Documentation:** Documenting the status of actions and the final risk assessment.

Risk is quantified as:

$$RPN = S \times O \times D$$

Where:

- **Severity (S):** Impact on the end-user. Note that this value typically remains constant unless the design concept changes completely.

- **Occurrence (O):** Probability of the cause occurring.
- **Detection (D):** Ability of current tests to find the failure before production.

THE RISK MATRIX

Risk Analysis

Step 4, 5, 6
Failure Action & Mitigation

Engine / Transmission System / Component Core Team			Prepared By			FMEA Date (Orig.)			FMEA Date (Update)		
Ref	Item (Element / Interface)	Elementary function	Potential Effect(s) of Failure	Potential Failure Mode	Potential Cause(s) Mechanism(s) of Failure	Drawing Characteristic and Design Specification	Current Design Controls Prevention (Design Criteria)	Current Design Controls Detection (Experimental Test)	Severity	Occurrence	Detection
R.P.N. = SEVERITY x OCCURRENCE x DETECTION											

The D-FMEA methodology | Artisan Model | 2021 December 31st

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Figure 68 Structure Analysis: Boundary Diagram example.

Beyond the 7 steps, the D-FMEA process requires continuous tracking to ensure that identified risks are actually mitigated. This is managed through specific Key Performance Indicators (KPIs):

- **Meeting Schedules:** Tracking adherence to the planned timeline for FMEA sessions to avoid delays in the product development plan.
- **Analysis Progress:** Monitoring the ratio of "Open" points (risks identified but not yet solved) versus "Closed" points (risks successfully mitigated).
- **Corrective Action Status:** Verifying that recommended actions have been implemented and validated experimentally.

Graphs and histograms are typically used to visualize the "To Do / Doing / Done" status, providing management with an immediate view of the reliability health of the project.

10.2.1 Case Study: IVECO Stralis MY24

The D-FMEA methodology was applied to the front bumper system of the IVECO Stralis truck. This case study highlights the "Package" challenge: integrating aesthetic/functional components without compromising safety sensors.

- **Problem:** The foldable step, used by drivers to clean the windshield, obstructed the field of view of the frontal radar when closed. This is a critical issue as the radar feeds the ADAS (Advanced Driver Assistance Systems).
- **Initial Risk:** High (RPN 126). Severity was 9 (Safety critical), Occurrence was 7.
- **Solution:** A geometric stack-up analysis was performed to verify the tolerance chain and alignment against supplier prescriptions.

- **Result:** The Occurrence dropped to 4, reducing the RPN to 72.
- **Problem:** The screws fixing the plastic step covers were prone to corrosion, leading to detachment risks over the vehicle's lifecycle.
- **Initial Risk:** RPN 112 (Severity 7, Occurrence 8).
- **Solution:** The screw specification was upgraded. The resistance requirement was increased from 720 hours to 1000 hours of salt spray testing (likely following ISO 9227 standards), ensuring durability in harsh environmental conditions.
- **Result:** Occurrence dropped from 8 to 3, lowering the RPN to 42.

10.2.2 Conclusion

The seminars effectively demonstrated that safety and reliability are not just compliance exercises but integral parts of the engineering design process. By combining ISO 26262 for electronic safety goals and D-FMEA for robust mechanical integration, engineers can achieve the "Zero Defect" target, ensuring both regulatory compliance and customer satisfaction.

11 **Pirelli experience on Vizzola track**

The experimental activity was conducted at the **Pirelli test track in Vizzola Ticino**. The session aimed to analyze the limit handling characteristics of a **Rear-Wheel Drive (RWD)** vehicle in a real-world environment.

To safely explore the vehicle's behavior at the limit of adhesion, the track surface was **artificially wetted**. This condition significantly reduced the friction coefficient, allowing the vehicle to reach tire saturation at lower, safer speeds.



Figure 69 Overview of the Pirelli Vizzola test track layout.

The track experience allowed for a direct assessment of:

- The vehicle's understeer gradient in quasi-steady-state conditions.
- The transition from stable to unstable equilibrium induced by longitudinal forces (traction/braking).
- The practical implications of the "Friction Circle" concept when maneuvering near the limit of adhesion.

11.1 **Steady-State Analysis: Constant Radius Steering Pad**

The first part of the experiment involved a **Constant Radius Steering Pad** maneuver. This test is fundamental for generating the "Handling Diagram," which relates the steering angle to lateral acceleration.

The driver followed a circular trajectory of a fixed radius, gradually increasing the forward speed. This "quasi-steady-state" approach allows for the decoupling of transient effects (like yaw inertia), isolating the tire-road interaction.

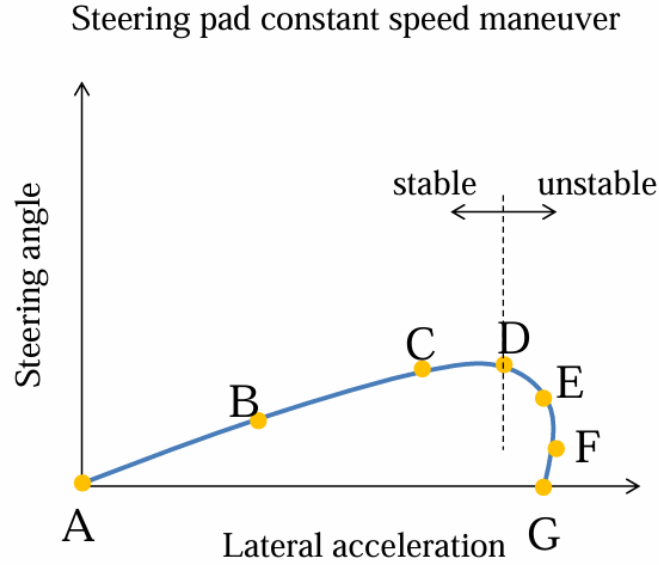


Figure 70 Theoretical Handling Diagram for the tested RWD vehicle, showing the transition from initial understeer to terminal oversteer.

The vehicle exhibited a mixed handling behavior, which is highly characteristic of powerful RWD setups:

- **Linear Region:** At low lateral accelerations, the relationship between steering angle and speed was linear, indicating a neutral or slightly understeering setup designed for general driving safety.
- **Initial Understeer:** As lateral acceleration increased, the vehicle initially exhibited a positive understeer gradient. The driver had to increase the steering angle to keep the car on the circular path, representing a stable equilibrium.
- **Limit Behavior (Terminal Oversteer):** Approaching the limit of adhesion, the handling diagram curve reversed its slope (negative gradient). The rear axle saturated before the front one, causing the vehicle's tail to slide outward. In this unstable region, the driver must reduce the steering angle (or counter-steer) to avoid a spin, showcasing the critical limit-handling nature of this RWD vehicle.

11.2 Transient and Combined Slip Dynamics

The second phase investigated how applying or removing torque alters the vehicle's balance. This section validates the theoretical **Friction Circle** (or Ellipse) concept, where the vector sum of longitudinal and lateral forces cannot exceed the maximum friction limit.

11.2.1 Power-On Maneuver (Limit Oversteer)

In this test, while cornering near the limit of adhesion, the driver aggressively applied full throttle (Power-On).

Physical Analysis:

1. **Load Transfer:** Acceleration causes a longitudinal load transfer from the front to the rear axle. Theoretically, increasing the vertical load on the rear tires should increase their lateral force capacity.
2. **Combined Slip Conflict:** However, the massive demand for tractive force to accelerate the RWD vehicle consumes the majority of the available friction at the rear contact patch. According to the friction circle, the available lateral force drops significantly.
3. **Result (Instability):** The rear axle saturates laterally before the front axle. The vehicle transitions into **Power Oversteer**, an unstable equilibrium condition where the yaw rate increases rapidly, potentially leading to a spin.

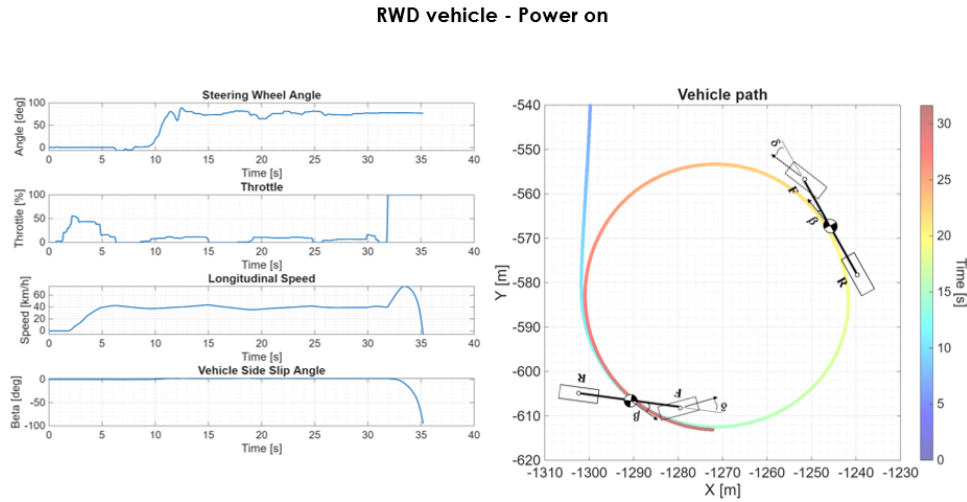


Figure 71 Effect of tractive force on lateral capacity. The resultant force vector exceeds the friction limit, reducing F_y capability.

11.2.2 Power-Off Maneuver (Tuck-in Effect)

Conversely, the driver performed a "Lift-off" maneuver, releasing the throttle mid-corner.

Physical Analysis:

1. **Load Transfer:** Deceleration causes a longitudinal load transfer from the rear to the front axle (pitching).
2. **Front Axle Response:** The increased vertical load on the front tires increases their cornering stiffness and lateral force capacity, as shown in the load sensitivity diagram.
3. **Rear Axle Response:** The rear tires are relieved of the tractive effort, allowing them to dedicate their entire friction budget to lateral forces, although their vertical load is reduced.
4. **Result (Tuck-in):** The dominant effect is the increased "bite" of the front axle. The vehicle tightens its trajectory, moving towards the inside of the curve. This phenomenon, known as **Tuck-in**, is generally stable and allows the driver to close the line during cornering without losing rear stability.

RWD vehicle - Power off

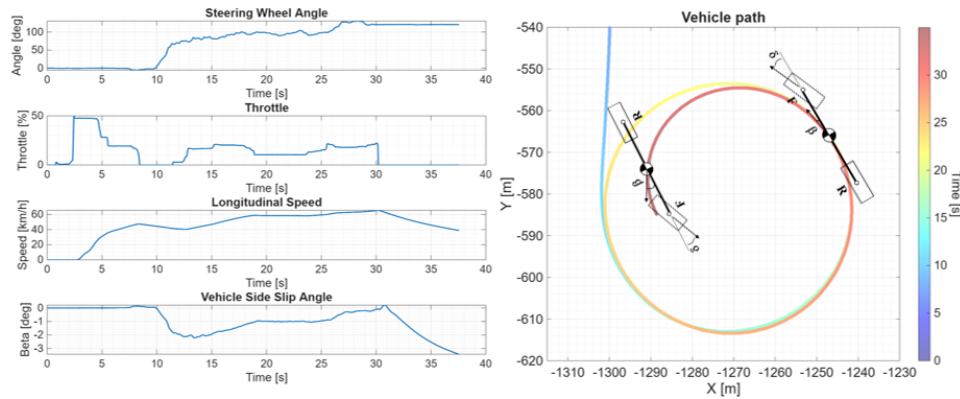


Figure 72 Modification of the lateral characteristic curve due to vertical load variation. Increased load on the front axle improves cornering stiffness.

11.3 Conclusion

The Vizzola experience provided empirical validation of the vehicle dynamics models discussed in the lectures:

1. **Stability vs. Control:** The steering pad tests confirmed that understeer is a stable, manageable condition, whereas the power-on tests demonstrated that oversteer is an unstable state requiring rapid counter-steering.
2. **Role of Load Transfer:** Handling is not a static property but changes dynamically. The "Tuck-in" effect during power-off is a direct consequence of longitudinal load transfer modifying the front/rear grip balance.
3. **RWD Characteristics:** The tests highlighted the sensitivity of RWD vehicles to throttle application. The interaction between longitudinal drive forces and lateral cornering forces (Combined Slip) is the deciding factor in the transition from stable cornering to oversteer.